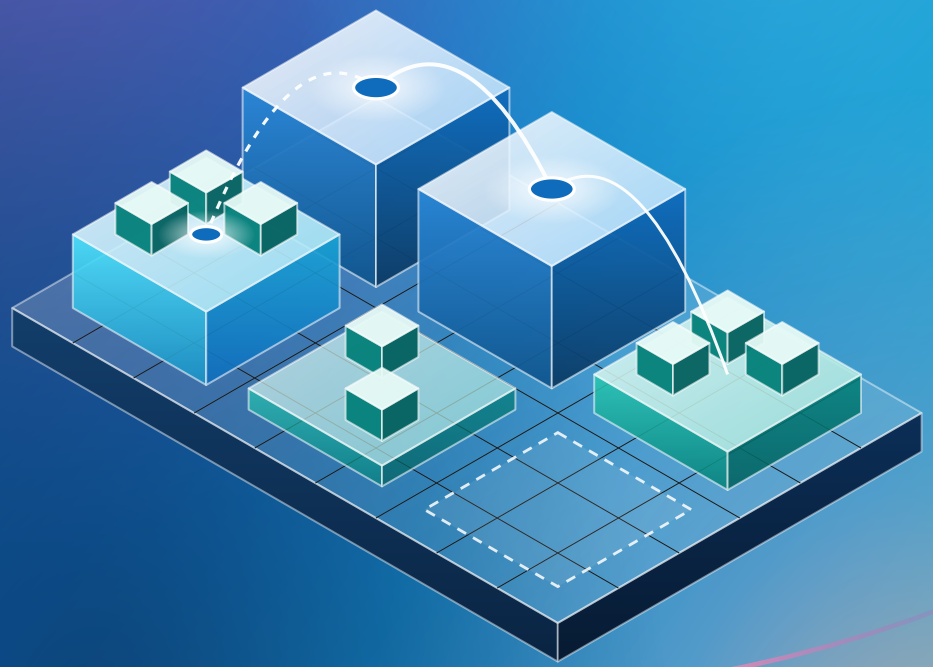
Microsoft

White paper

AZURE ARCHITECTURE PLATFORM STRATEGY

The adaptable **multi-region** Azure platform

Extending Azure landing zones for adaptable regional placement — qualifying regions, planning by workload archetype, and selecting the most efficient regional platform connectivity.



THREE PLANNING CONSTRUCTS

- Regional Platform Connectivity Profiles
- Workload Landing Zone Archetypes
- Regional Qualification

October 2026

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MULTI-REGION PLATFORM WHITEPAPER



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EXECUTIVE SUMMARY

Make regional choice a governed platform capability — not a one-time prediction

Adding an Azure region shouldn't mean starting another landing-zone project. If you already run Azure landing zones, most of what a new region needs already exists: the governance, identity, policy, automation, and operating model. This paper shows how to reuse that foundation so that additional regions become **qualified options that workloads can use selectively**, without creating separate governance models or reproducing an identical platform footprint in every region.

> The objective is not to prebuild every capability everywhere. It is to keep one operating model, qualify regional options against known business and workload requirements, and enable only the platform capabilities that each region needs.

What adding a region involves

A new region is often assumed to be a large project. With this approach, it is a short series of decisions and a targeted change to the platform you already run.

COMMON ASSUMPTION	WITH THIS APPROACH
✕ **A new region needs its own landing-zone design.**	✓ Reuse the existing landing zone. The region is added to the policies, automation, and monitoring you already run. Framework overview
✕ **Every region needs a full hub.**	✓ Start with the most efficient connectivity profile that meets the requirements. Two of the four profiles need no local hub. Part 1
✕ **Every region needs another ExpressRoute circuit.**	✓ Reuse existing hybrid connectivity where reachability and resiliency requirements allow. Add circuits only when a requirement justifies them. Part 3
✕ **Enabling a region means migrating workloads.**	✓ Nothing has to move. Workloads use the region when their own requirements call for it. Part 5
✕ **Qualification is a long assessment.**	✓ It is four checks, each with one question, answered from what your teams already know. Unknowns are recorded and revisited, not treated as blockers. Part 2

The first additional region takes the most thought, because the decisions are made for the first time. Later regions reuse those decisions and become mostly configuration. For more examples, see Common misconceptions (Putting it into practice).

EXECUTIVE SUMMARY · CONTINUED

How it works

The framework has five parts. Each part answers one question.

1	**Understand concepts**	**What is a multi-region platform, what can a region provide, and what do applications need?**	Four connectivity profiles describe the platform options. Four archetypes describe what applications require. Part 1
2	**Qualify regions**	**Can this region support the workloads we expect?**	Four checks lead to one of four outcomes: excluded, candidate, conditional, or qualified. Part 2
3	**Select the regional design**	**What should the platform provide in a qualified region?**	Make three choices from short lists: the connectivity profile, where each shared service is provided, and whether the hybrid connectivity you already have is reused. The region is then **selected**. Part 3
4	**Prepare the region**	**What has to exist before workloads arrive?**	Build only what the selected design calls for, through the automation you already run. Two of the four profiles need no hub. The region is then **ready for workloads**. Part 4
5	**Place workloads**	**Should this workload use the region now?**	Decide per workload, with current evidence. The region is then **in use**. Part 5

The approach at a glance

Extend, do not replicate

The landing-zone operating model stays consistent across the Azure estate. New regions add only the **most efficient platform capabilities** required by the workloads expected to use them, rather than replicating an existing regional footprint.

Qualify regions against defined requirements

Regional qualification evaluates business and geographic needs, data and compliance constraints, workload requirements and dependencies, and regional capabilities. A region can remain excluded, conditional, or qualified based on documented evidence, constraints, owners, and revalidation triggers.

Qualification creates a **governed regional option**; it does not permanently approve the region for every workload.

Plan by workload requirements, not individual applications

Workload Landing Zone archetypes provide a planning abstraction for recurring connectivity and dependency requirements. They help platform teams anticipate what future workloads may require without attempting to predict every application in advance.

Archetypes describe **what applications require**. Regional connectivity profiles describe **what the platform provides**. The relationship is intentionally not one-to-one.

EXECUTIVE SUMMARY · CONTINUED

Choose the most efficient regional connectivity

Regional Platform Connectivity Profiles range from **Disconnected Spokes** through **Remote Hub Connected** and **Minimal Regional Hub** to a **Full Regional Hub**.

These profiles are not maturity stages. Select the **most efficient profile** for current workload and platform requirements. Add local capabilities only when a measurable need for connectivity, latency, resiliency, security, data handling, or operations justifies them.

Strategic geographic hubs and shared services can continue to support other regions where dependencies and service characteristics permit, avoiding unnecessary duplication.

Keep platform readiness and workload placement separate

Regional qualification determines whether a region can become a viable platform option. Actual workload placement occurs later, when a specific workload and its current requirements are known.

During onboarding, validate current Azure service and SKU availability, regional access, quota, dependencies, connectivity, and resiliency requirements. Placement, relocation, recovery, and multi-region operation remain **workload-specific decisions**.

New workload findings can also feed back into regional qualification and trigger platform expansion, additional validation, or reassessment.

Keep the operating model consistent while allowing regional adaptation

The operating model stays consistent across regions; implementation can vary where business, workload, or regulatory requirements call for it.

Workload placement, connectivity profiles, local shared-service footprints, Azure services and SKUs, resiliency choices, and other regional parameters can adapt to current requirements.

Region-agnostic does not mean region-random. Regional choice operates within a governed framework of qualified options, documented constraints, and current evidence.

New to the framework? Getting started (Putting it into practice) walks through the first additional region in six steps, and the decision trees (Parts 2 and 3) put the two main decisions on one page.

THE GOAL

The goal is not to select the perfect two regions. **It is to make additional regions significantly easier to adopt when business and workload requirements justify them.**

FRAMEWORK OVERVIEW

Extending Azure landing zones for adaptable regional placement

The same operating model applies across the estate, while each region is **qualified against the workloads** it may need to support.

Azure landing zones provide a governed foundation for identity, connectivity, security, management, and application delivery. Many organizations initially establish this foundation around a small, fixed set of Azure regions — typically a primary region and a disaster-recovery region, reflecting traditional datacenter and colocation design. This model works while demand and placement requirements remain predictable. It becomes less adaptable as business expansion, data boundaries, service and AI availability, latency requirements, and other regional conditions evolve.

Because platform teams cannot predict every future workload, this paper uses three planning constructs: Regional Platform Connectivity Profiles, Workload Landing Zone Archetypes, and Regional Qualification.

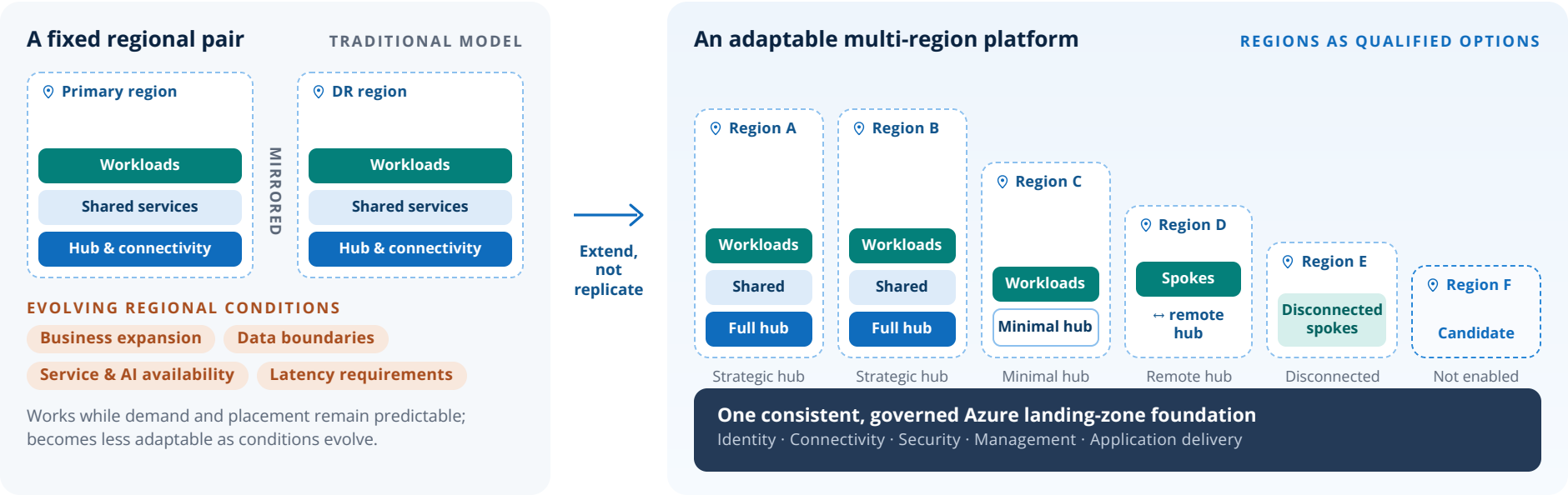


FIGURE 1 **From a fixed regional pair to governed regional options.** Instead of mirroring a complete footprint in each new region, the platform extends one governed foundation and gives each region only the capability its workloads require. Regional capability is a spectrum, not a copy.

FRAMEWORK OVERVIEW · WHAT STARTS THE CONVERSATION

Common drivers of multi-region platform discussions

Multi-region platform work rarely starts as an architecture initiative. It usually starts with a change in the business, regulation, workload requirements, service availability, or risk. These drivers open the discussion; they do not determine the regional architecture or workload-placement outcome.

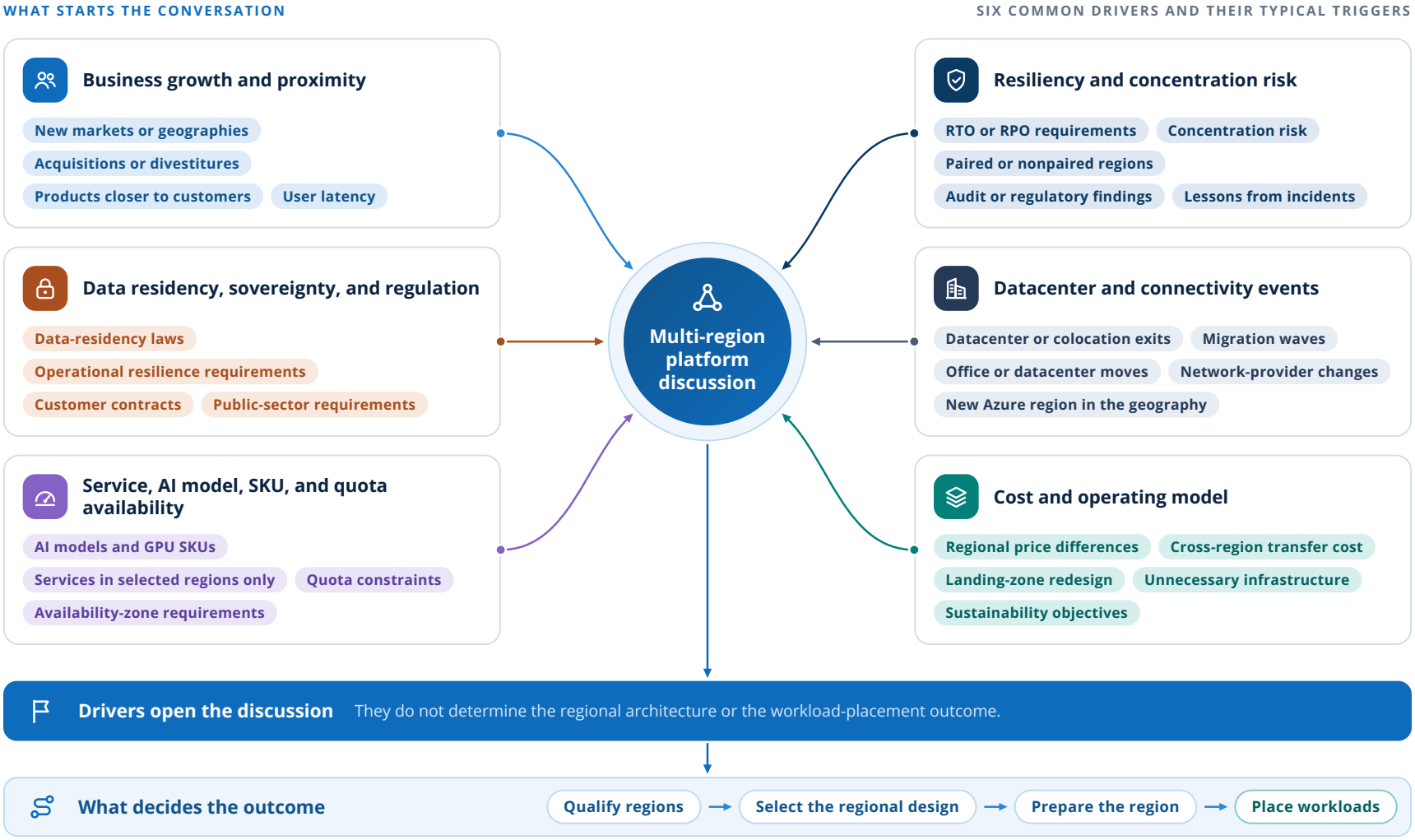


FIGURE 2 **Common drivers of multi-region platform discussions.** Typical patterns observed across enterprise estates; each one opens a discussion rather than determining a predetermined outcome.

SCOPE OF THIS PAPER

Azure public-cloud regions. Sovereign cloud offerings and Azure Local have their own constraints and are not covered, although the same qualification logic can inform them.

THREE PLANNING CONSTRUCTS

Balancing advance planning with flexibility

Platform teams cannot predict every future workload. The three constructs let the platform plan for what is known, stay flexible for what is not, and hand workloads a set of governed regional options to choose from.

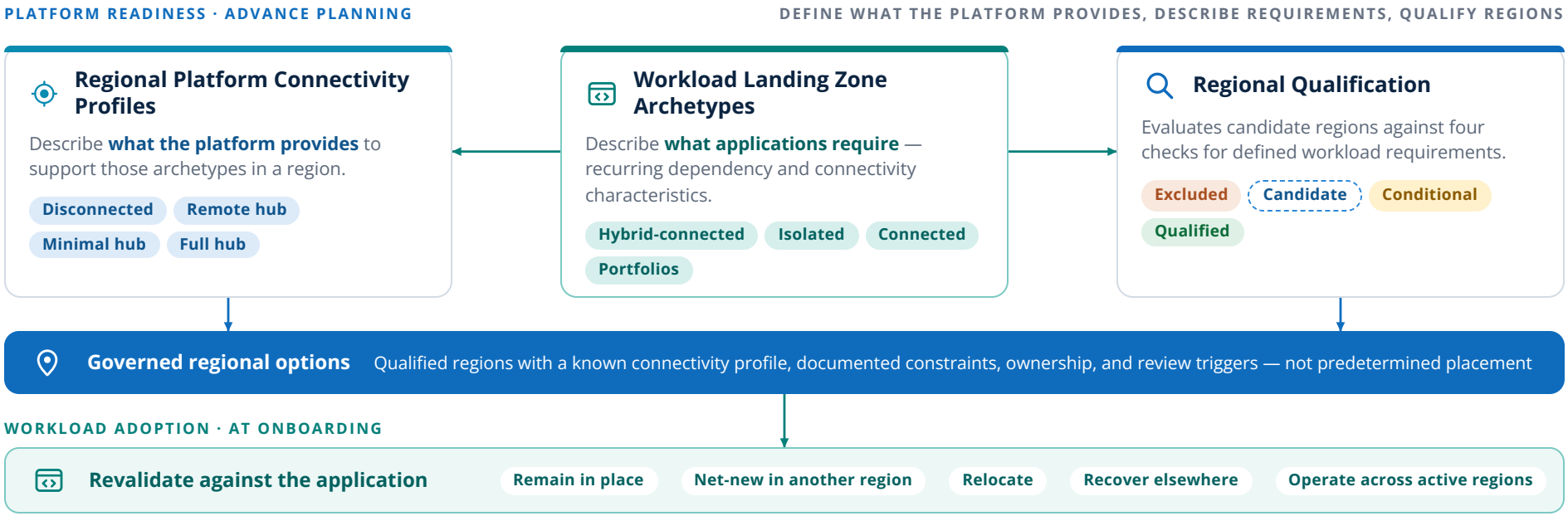


FIGURE 3 **How the three constructs connect.** Archetypes (application requirements) and connectivity profiles (platform capability) stay distinct; together with regional qualification they produce governed options. Workloads adopt those options only after onboarding revalidation — migration and resiliency remain workload-specific decisions.

REGIONAL PLATFORM CONNECTIVITY PROFILES

Describe what the platform provides in a region: Disconnected Spokes, Remote Hub Connected, Minimal Regional Hub, or Full Regional Hub. They are not maturity stages; the most efficient profile can remain the target as long as it continues to satisfy requirements.

WORKLOAD LANDING ZONE ARCHETYPES

Group recurring dependency and connectivity characteristics so platform teams can plan before every workload is known. The four archetypes are *Hybrid-Connected Applications*, *Isolated Cloud-Native or AI Applications*, *Connected Cloud-Native or AI Applications*, and *Interconnected Application Portfolios*.

REGIONAL QUALIFICATION

Provides a repeatable way to assess candidate regions. Four checks lead to an excluded, candidate, conditional, or qualified regional option, backed by current evidence, known constraints, ownership, and review triggers.

PROGRESSIVE ADOPTION JOURNEY

Establish regional options first — then use and expand them as workload needs justify

Together, these constructs create governed regional options without predetermining workload placement. During onboarding, teams revalidate the region against the workload's current dependencies and requirements, then decide whether to stay put, deploy net-new, relocate, recover elsewhere, or operate across regions.

The journey below shows the recommended sequence for adopting regional options. It is **not a mandatory maturity path**; platform capabilities and workload use evolve only when business and application requirements justify them.

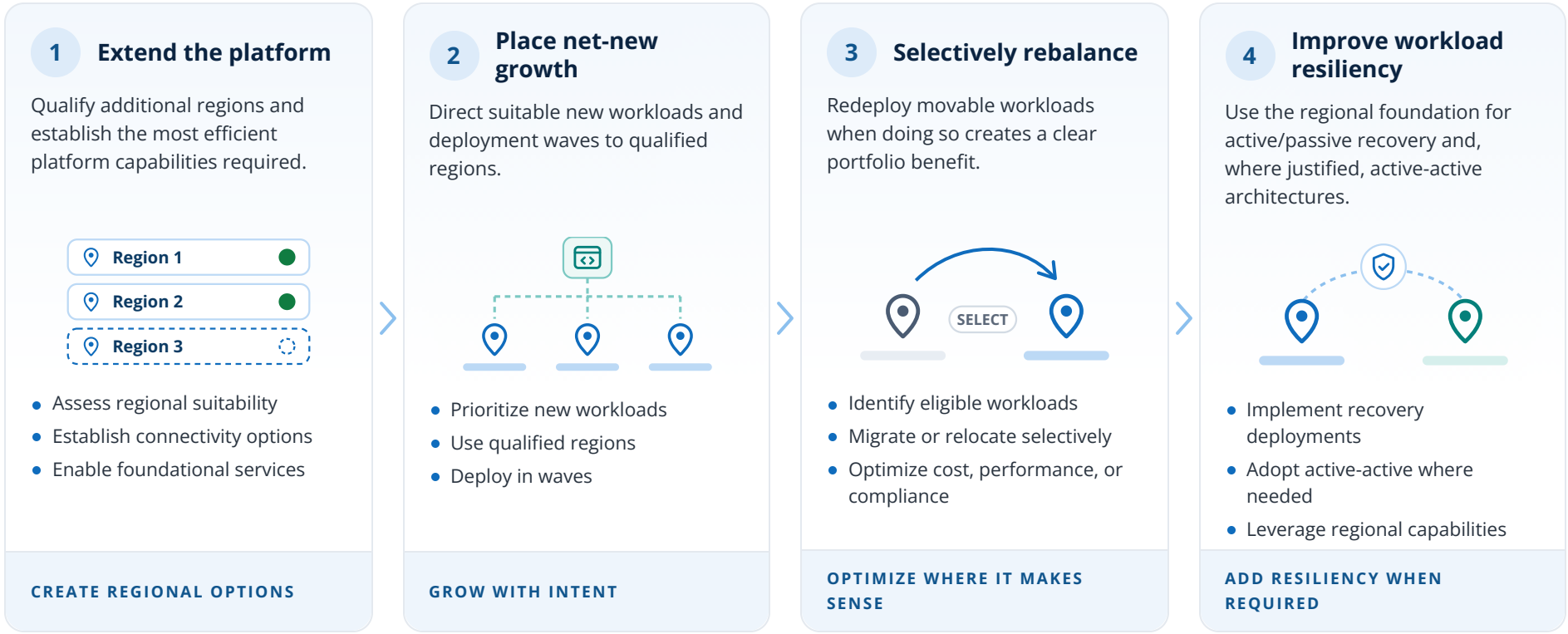


FIGURE 4 **A progressive journey for multi-region adoption.** Regional options are created first and consumed selectively. Each step is justified by business and application need; a platform can stop at any step that continues to meet requirements.

UNDERSTAND CONCEPTS · TWO LAYERS, TWO DECISIONS

01 Multi-region platform and multi-region workload

The **platform** creates governed regional choices for the estate; the **workload** applies one or more of those choices to a specific application.

A multi-region platform and a multi-region workload operate at different layers. The platform creates governed regional choices for the customer estate; the workload applies one or more of those choices to a specific workload. Keeping these concepts separate prevents multi-region strategy from being treated as an automatic requirement for a new landing zone, workload migration, active-active high availability, or active/passive disaster recovery.

PLATFORM READINESS DOES NOT IMPLY

- × A new landing zone
- × Workload migration
- × Active-active HA
- × Active/passive DR

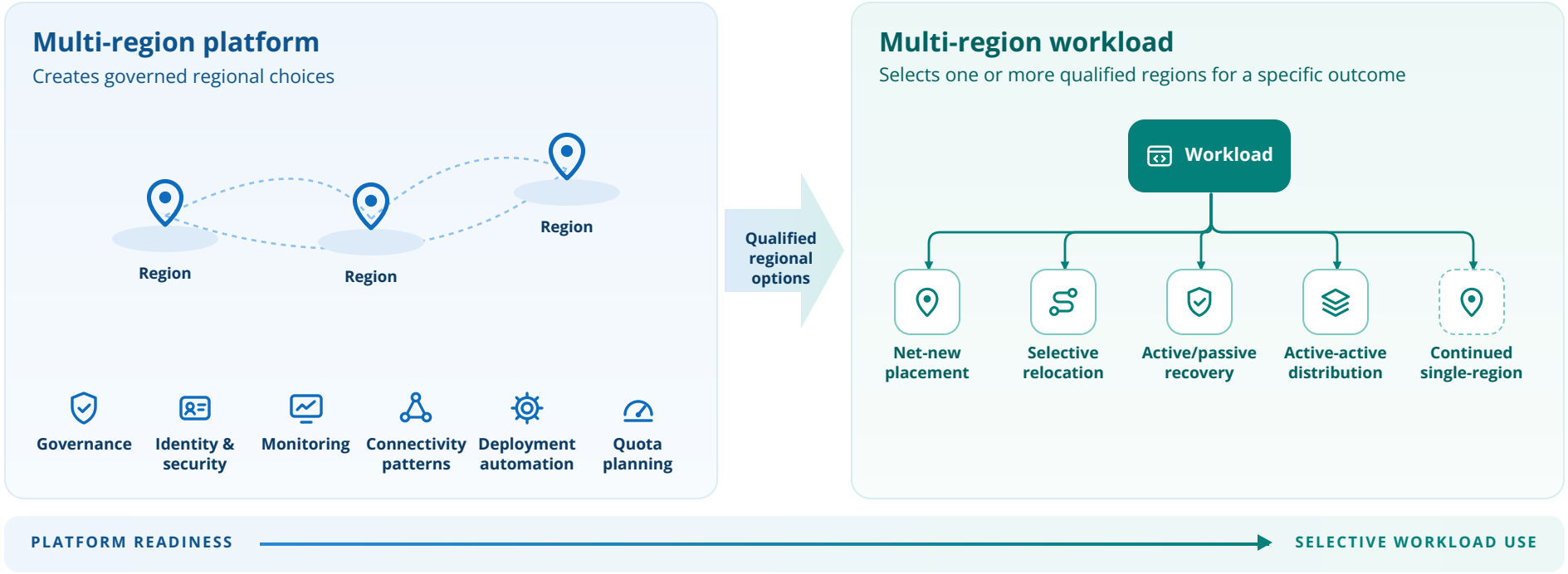


FIGURE 5 **Multi-region platform and multi-region workload operate at different layers.** Platform capabilities make regions available as qualified options; each workload then selects its own outcome — which can legitimately be continued operation in a single region.

Two layers compared

TABLE 1

	MULTI-REGION PLATFORM	MULTI-REGION WORKLOAD
Purpose	Enable placement flexibility across qualified Azure regions	Meet **workload-specific objectives** and requirements such as availability, disaster recovery, user proximity, or regulatory requirements
Focus	Regional readiness and consistency of shared platform capabilities	Application architecture, data, dependencies, traffic, availability, and recovery
Design driven by	The needs of the portfolio of workloads the platform is expected to support	The specific workload's requirements, such as availability, resiliency, RTO/RPO, latency, or data requirements
Outcome	Additional regions qualified through the existing landing-zone operating model, approved connectivity patterns, deployment automation, and quota planning	Net-new placement in another region, selective relocation, active/passive recovery, active-active distribution, or continued operation in one region

DEFAULT POSTURE

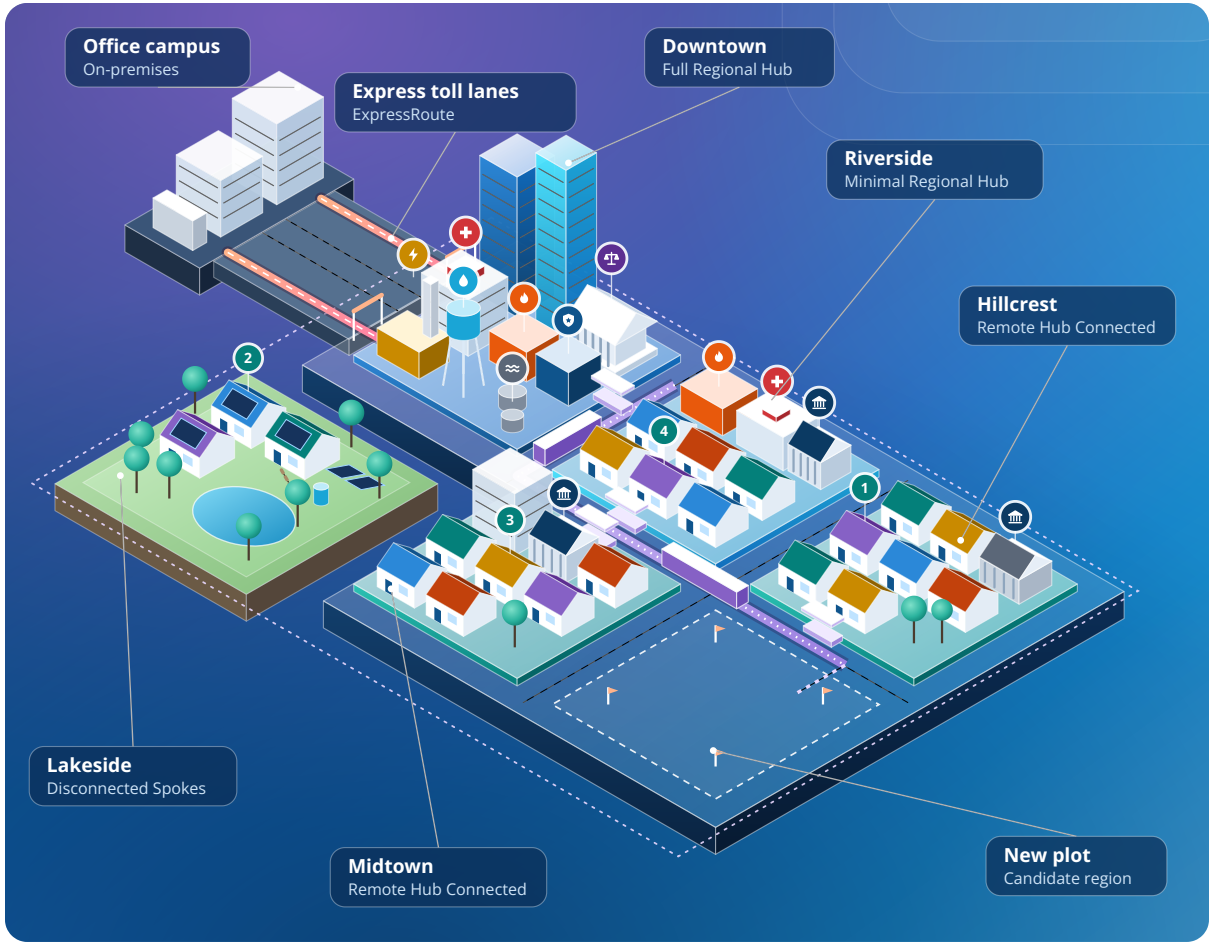
Reuse the existing Azure landing-zone operating model by default. Extend regional policy parameters, connectivity, shared services, automation, and operations only where documented workload requirements justify them.

A multi-region platform doesn't require a separate connectivity landing zone or identical shared services in every region; **workload requirements determine the cross-region design**.

PART 1 · AN EVERYDAY ANALOGY

The platform is the city. Workloads are the people and businesses that live in it.

The city builds neighborhoods, services, roads, transit, and laws once. A family settles in one neighborhood; a bank spreads across several. Neither builds a new city.



NUMBERED PINS — **SINGLE-REGION WORKLOADS**

1. **Hybrid-connected** Hillcrest
2. **Isolated** Lakeside
3. **Connected** Midtown
4. **Interconnected portfolio** Riverside

BANK — **MULTI-REGION WORKLOAD**

Branches in **Riverside** and **Midtown** serve customers together. A backup operations center in **Hillcrest** stands by.

PLACES — **REGIONS AND ON-PREMISES**

Neighborhood
An **Azure region**. The label names its connectivity profile.

Office campus
On-premises datacenters and users, outside the city.

Staked plot
A **candidate region**: zoned, but not built yet.

ROUTES — **CONNECTIVITY**

Express toll lanes
ExpressRoute. Two interchanges give the campus two separate paths into the city.

Transit line and stations
Private connectivity between regions. Hillcrest and Midtown ride it to Downtown.

LAWS — **GOVERNANCE AND POLICY**

City hall
Laws, codes, and zoning: **governance and policy**, set once for the whole city.

City limits
Where the laws apply: every neighborhood, Lakeside included.

CITY SERVICES — **SHARED SERVICES**

- **Hospital or clinic**
- **Fire station**
- **Police station**
- **Power plant**
- **Water tower**
- **Sewer plant**

Together they are the platform's **shared services**. Downtown has every one. Riverside has a fire station and a clinic. Hillcrest and Midtown use Downtown's. Lakeside needs none.

FIGURE 6 **The platform as a city.** Each neighborhood is a region with its own connectivity profile. The legend explains each icon on the map.

01

UNDERSTAND CONCEPTS · WHAT THE PLATFORM PROVIDES

Regional Platform Connectivity Profiles

Choose the most efficient profile that supports the workloads expected to use the region.

Enabling an additional Azure region doesn't require changing or duplicating the full platform footprint of an existing geographic hub. An organization can maintain a small number of strategic geographic hubs that provide resilient hybrid connectivity and common enterprise services. Other Azure regions can use a more efficient profile and consume capabilities from those hubs when workload requirements permit.

The four profiles form a spectrum: from Disconnected Spokes, with no dependency on a geographic hub, through Remote Hub Connected and Minimal Regional Hub, to a Full Regional Hub with greater local capability and independence. They **aren't maturity stages**. A profile can remain the right target for a region as long as it continues to satisfy workload and platform requirements.

Profiles describe platform capabilities, not workload designs; Workload Landing Zone archetypes describe what applications need.

NO DEPENDENCY ON A GEOGRAPHIC HUB → GREATER LOCAL CAPABILITY AND INDEPENDENCE

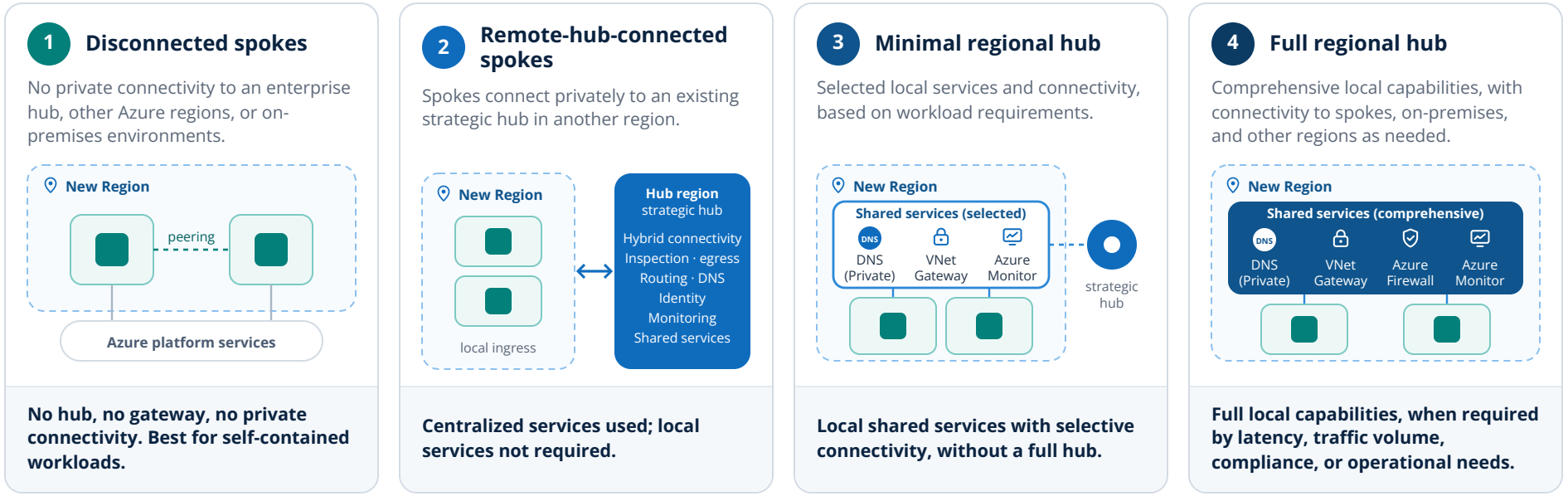


FIGURE 7 **The spectrum of regional platform connectivity profiles.** Moving right adds local capability and independence — and cost and operational scope. Services shown are examples; each hub includes only what its workloads require. Any profile can be the right long-term target.

1 Disconnected spokes



PROFILE 1 OF 4

Workload spokes operate without private connectivity to an enterprise geographic hub, another Azure region, or an on-premises environment. They can still use direct connectivity to other spokes where required and approved private or public connectivity to Azure platform services.

FITS WHEN	Workloads can operate without private connectivity to an enterprise geographic hub, another Azure region, or an on-premises environment, and their required platform services can be consumed locally or through approved Azure service connectivity. This can include self-contained cloud-native, AI, development, test, batch, HPC, VDI, or internet-facing workloads where those conditions are met.
REGIONAL FOOTPRINT	Workload resources plus the controls needed to deploy, secure, monitor, and operate them. These spokes still use the estate's standard identity, policy, security, and deployment practices.
KEY CONSIDERATIONS	Validate administration, identity, DNS, secrets, telemetry, data movement, software distribution, private endpoints, incident response, and support procedures. Where applications require east-west communication within the region, evaluate direct spoke-to-spoke connectivity without assuming that a regional hub is required. Confirm that no hidden dependency requires connectivity to a geographic hub or on-premises environment.
REASSESS THE PROFILE WHEN	Workloads require private access to enterprise services, centralized security inspection, on-premises systems, another application portfolio, or capabilities hosted through a geographic hub.

2 Remote-hub-connected spokes



PROFILE 2 OF 4

Workload spokes in the additional region use cross-region connectivity to access an existing strategic geographic hub. The geographic hub can provide hybrid connectivity, security inspection, internet egress, routing, DNS, identity, monitoring, or other shared services without requiring a complete hub to be deployed locally. Cross-region connectivity can be implemented through appropriate Azure networking capabilities based on the required topology and routing model. Public application ingress can remain local through services such as Azure Front Door or Application Gateway when appropriate.

FITS WHEN	Workloads require private access to selected enterprise or on-premises services but can tolerate the latency, availability coupling, data-transfer implications, and dependency on connectivity to another region.
REGIONAL FOOTPRINT	Workload spokes and any capabilities that must remain close to the workloads. Gateways, firewalls, hybrid connectivity, routing services, and selected shared services can remain in an existing geographic hub when workload requirements permit.
KEY CONSIDERATIONS	Measure latency and bandwidth using representative traffic. Evaluate cross-region data-transfer cost, routing and transit behavior, inspection paths, geographic hub scale, dependency criticality, and operational ownership. Document how workloads behave if the geographic hub or cross-region connectivity is degraded or unavailable.
REASSESS THE PROFILE WHEN	Cross-region traffic, latency, east-west communication, dependency criticality, regulatory requirements, scale, or availability coupling justify deploying selected connectivity or shared-service capabilities locally.

3 Minimal regional hub



PROFILE 3 OF 4

The additional region includes a limited hub footprint that provides selected local connectivity and shared services while continuing to consume other capabilities from an existing geographic hub. The minimal hub provides only the local capabilities required by its workloads rather than replicating a complete platform footprint. Depending on workload requirements, these can include east-west routing, DNS, monitoring, backup, security inspection, egress, private connectivity, selected identity or directory services, on-premises connectivity, or other services that must remain close to the workloads.

FITS WHEN	Workloads require some local shared services, application-to-application connectivity, lower-latency dependency paths, regional security controls, or greater independence than a remote-hub-connected profile can provide.
REGIONAL FOOTPRINT	A regional hub containing only the connectivity and shared-service capabilities that must operate locally. Hybrid connectivity, specialized networking functions, central management, and other services can remain in a strategic geographic hub when appropriate.
KEY CONSIDERATIONS	Clearly define which capabilities operate locally and which remain remote. Avoid ambiguous routing, overlapping responsibilities, and unnecessary duplication of platform controls. Validate workload and operational behavior when either local capabilities or remote dependencies are unavailable. When on-premises connectivity is required, determine whether existing ExpressRoute circuits and gateway topology can provide the required reachability before introducing additional circuits; **do not assume that each Azure region requires a dedicated ExpressRoute circuit**.
REASSESS THE PROFILE WHEN	Workload growth introduces additional local connectivity or shared-service requirements, multiple interconnected application portfolios require broader regional capabilities, stricter independence requirements emerge, significant hybrid connectivity is required locally, or other requirements justify expanding the regional footprint.

4 Full regional hub



PROFILE 4 OF 4

The region provides the complete set of local connectivity and shared services required for its designated workloads and geographic responsibilities. Depending on those requirements, this can include local hybrid connectivity, security inspection, routing, internet ingress and egress, DNS, identity services, monitoring, backup, key management, and other required platform capabilities.

A full regional hub uses the same operating model and control baselines as other strategic hubs. **It does not have to be an identical copy**; its capabilities should reflect the workloads and geography it supports.

FITS WHEN	Workload and platform requirements justify a comprehensive local capability set — for example, local hybrid connectivity, substantial local shared services, strict latency or data-boundary controls, significant interconnected application portfolios, or sufficient independence to continue operating when another region or geographic hub is unavailable.
REGIONAL FOOTPRINT	The approved set of local connectivity, shared services, operational capabilities, and independent critical paths required by the region's workload portfolio.
KEY CONSIDERATIONS	Confirm that the required level of independence justifies the additional cost and operational scope. Validate automation, observability, support ownership, security operations, configuration consistency, dependency behavior, and lifecycle management. Where on-premises connectivity is required, determine whether existing ExpressRoute circuits can provide the required reachability and resiliency before introducing additional connectivity.
REASSESS THE PROFILE WHEN	Workload demand or requirements decrease, services can be safely centralized, dependencies change, or the full regional footprint no longer provides sufficient value relative to its cost and operational complexity.

01

UNDERSTAND CONCEPTS · WHAT APPLICATIONS REQUIRE

Workload Landing Zone Archetypes

A planning abstraction that groups recurring application connectivity and dependency requirements.

An archetype describes what an application or application portfolio requires from the platform. It doesn't replace the existing Azure landing-zone hierarchy.

Assign each workload the closest-fitting archetype and record any material requirements that fall outside it. Portfolio-level dependencies can also affect the platform capabilities needed across several workloads.

Use archetypes during regional qualification to anticipate the capabilities a candidate region may need to support. Where future workload requirements aren't yet defined, they give you planning guardrails without constraining later placement, which happens once the workload's current requirements, regional capabilities, and platform readiness are known.

1 Hybrid-Connected Applications



Significant dependencies on on-premises environments, enterprise networks, or centralized shared services that require persistent private connectivity.

- Persistent private connectivity
- Profile set by dependency latency, resiliency, regulation, operations
- Common in migrations and hybrid scenarios

TYPICAL CONNECTIVITY PROFILES

Remote hub | Minimal hub | Full hub

2 Isolated Cloud-Native or AI Applications



Self-contained workloads that can operate without persistent private connectivity to enterprise, on-premises, or centralized shared-service dependencies.

- No persistent private dependencies
- Can often use disconnected spokes
- Flexible placement across regions

TYPICAL CONNECTIVITY PROFILES

Disconnected spokes

Connected, where justified

3 Connected Cloud-Native or AI Applications



Primarily self-contained cloud-native or AI applications that require private connectivity to selected enterprise services, data sources, APIs, or other Azure environments.

- Private access to selected dependencies
- Profile set by latency, traffic, resiliency, security
- Common for data, analytics, and AI platforms

TYPICAL CONNECTIVITY PROFILES

Remote hub | Minimal hub | Full hub

4 Interconnected Application Portfolios



Groups of applications with meaningful dependencies on each other — shared data, common services, east-west communication, or coordinated connectivity and operations.

- Portfolio-level archetype
- Aggregate requirements set the profile
- Direct spoke-to-spoke or a regional hub for east-west traffic

TYPICAL CONNECTIVITY PROFILES

Remote hub | Minimal hub | Full hub

FIGURE 8 **Four Workload Landing Zone archetypes.** Grouped by the platform connectivity and dependency capabilities each requires. See Figure 9 for profile relationships.

HOW ARCHETYPES MAP TO CONNECTIVITY PROFILES

Archetypes describe what applications require. Profiles describe what the platform provides.

An archetype doesn't dictate a profile: one archetype can use different profiles, and one profile can serve several archetypes. Use the archetypes expected in a region to understand what it needs, then choose the most efficient profile that meets those needs. For example, a hybrid-connected application might run well on a Remote Hub Connected profile in one region and need a Minimal Regional Hub in another, depending on latency, traffic, and resiliency requirements.

ARCHETYPES WHAT APPLICATIONS REQUIRE	REGIONAL PLATFORM CONNECTIVITY PROFILES · WHAT THE PLATFORM PROVIDES Disconnected spokes	Remote-hub-connected spokes	Minimal regional hub	Full regional hub	WHAT DRIVES THE CHOICE
1 **Hybrid-Connected Applications**	—	● Typical	● Typical	● Typical	Dependency latency, resiliency, regulatory, operational, and shared-service requirements.
2 **Isolated Cloud-Native or AI Applications**	● Often	○ Where justified	○ Where justified	○ Where justified	Often disconnected spokes. A connected profile where private management, security, data access, or centralized services justify connectivity.
3 **Connected Cloud-Native or AI Applications**	—	● Typical	● Typical	● Typical	Private connectivity to selected dependencies; dependency latency, traffic patterns, resiliency, security, and operational requirements decide which profile.
4 **Interconnected Application Portfolios**	○ Spoke-to-spoke	○ Where acceptable	○ When justified	○ When justified	Aggregate portfolio connectivity, shared-service, resiliency, scale, and operational requirements.

● Typical fit ○ When specific requirements justify it — Not indicated as typical

FIGURE 9 **Archetype-to-profile relationships are many-to-many.** Each row is a set of options, not an assignment; the selected profile follows dependencies, latency, traffic, resiliency, and independence requirements.

Interconnected portfolios: east-west traffic

Where significant east-west traffic exists within a region, evaluate whether direct spoke-to-spoke connectivity or a regional hub is a better path than routing through a remote hub. Direct connectivity may suit a few tightly coupled workloads; a Minimal or Full Regional Hub fits broader routing, inspection, resiliency, or operational needs.

Portfolio-level requirements

Because a portfolio's combined dependencies can require capabilities beyond any single application, select its profile from aggregate connectivity, shared-service, resiliency, scale, and operational requirements — not from a single workload.

HOW ARCHETYPES DIFFER FROM OTHER ARCHITECTURE TERMS

Four architecture concepts — related, but not interchangeable

Several architecture concepts help identify and organize workload requirements, but they describe different aspects of a workload and should not be used interchangeably. An archetype is the only one of them that describes what a workload requires from the platform.

TERM	WHAT IT DESCRIBES	EXAMPLES	LEVEL
Architecture style	The fundamental structural organization of an application	Microservices, N-tier, event-driven, web-queue-worker, big compute	**Application architecture**
Architecture / design pattern	A reusable solution to a recurring technical problem	Circuit Breaker, Retry, CQRS, Strangler Fig, Competing Consumers	**Design technique**
Workload pattern	A recurring workload behavior or set of operational characteristics that can influence platform requirements	Latency-sensitive, data-intensive, batch-oriented, globally distributed, hybrid-dependent	**Workload characteristics**
Workload Landing Zone archetype	A classification of a workload based on the platform connectivity and dependency capabilities it requires	Hybrid-Connected Applications, Isolated Cloud-Native or AI Applications, Connected Cloud-Native or AI Applications, Interconnected Application Portfolios	**Platform-consumption model**

FIGURE 10 **Architecture concepts used to organize workload requirements.** Only the Workload Landing Zone archetype describes what a workload needs from the platform; it is the concept used in regional qualification.

NOT THE BUILT-IN LANDING-ZONE ARCHETYPES

The Azure landing zones reference architecture also uses *archetypes*: built-in definitions of what must be true for a landing zone to meet environment and compliance requirements at a given scope. Workload Landing Zone archetypes in this paper describe what a workload requires from the platform — a different concept.

ONE ARCHETYPE PER WORKLOAD

Classify each workload by its closest-fitting archetype. Record requirements beyond it as explicit additions to that archetype, rather than forcing the workload into multiple classifications.

REFERENCE LINKS

› Architecture styles › Built-in Azure landing zone archetypes

02

QUALIFY REGIONS

Regional qualification

Which Azure regions should be **prepared** to support workload requirements — not whether an individual workload should use multiple regions.

Regional qualification determines which Azure regions should be prepared to support workload requirements. It does not determine whether an individual workload should use multiple regions or define its multi-region architecture. Those are workload-specific design decisions. Use the following four checks to evaluate candidate regions consistently.

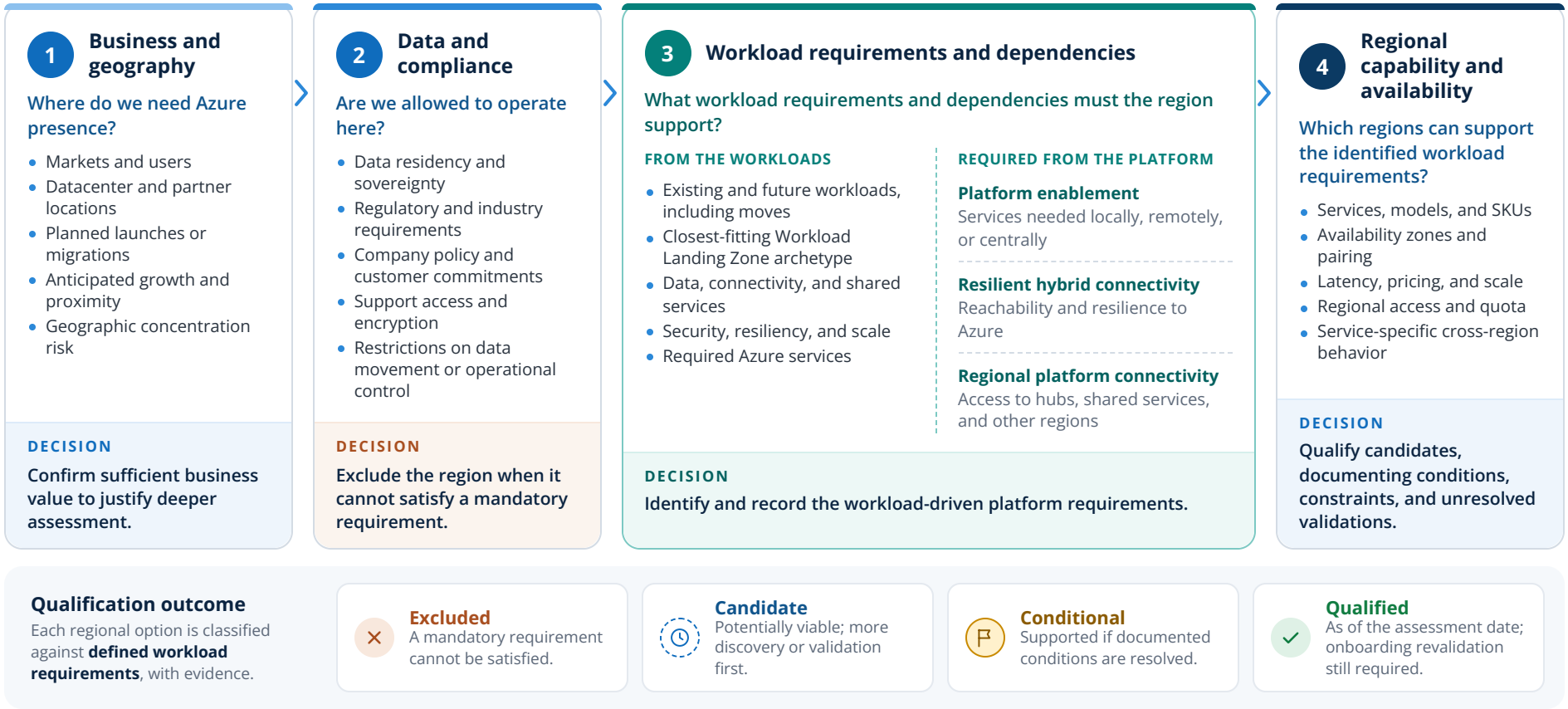


FIGURE 11 **Four checks — the recommended sequence.** Outcomes are specific to defined workload requirements, never a blanket certification; workload resiliency requirements are identified during workload design.

A region that meets all four checks is **qualified** for the defined workload requirements as of the assessment date. The next two pages define each outcome and walk the four checks as a single decision tree.

REGIONAL ASSESSMENT OUTCOME

Classify regional options, with evidence

Consolidate the workload requirements and platform needs identified through the four checks. Assess each candidate region against the same criteria and document the result, key tradeoffs, and outstanding validation. Reuse the assessment in architecture reviews and regional planning.

Classify each regional option as:

Excluded
A mandatory requirement cannot be satisfied.

Candidate
The region is potentially viable, but additional discovery or validation is required before qualification.

Conditional
Current evidence supports the defined workload requirements if one or more documented conditions are resolved.

Qualified as of the assessment date
Current evidence supports the defined workload requirements, platform ownership is established, and workload-onboarding revalidation is still required.

Record the evidence date, review date, workload requirements in scope, key constraints, outstanding validations, owner, and revalidation triggers. Add the platform connectivity profile when you select it in Part 3. Qualification is a dated planning input, not permanent approval; decision-time validation at onboarding remains authoritative.

A REPEATABLE INPUT, NOT A ONE-TIME EXERCISE

Assess each candidate against consistent criteria → **Classify** against requirements, with evidence → **Inform** architecture reviews and regional planning

Workload onboarding revalidates options and feeds new findings back into the assessment.

Illustrative: one portfolio, classified per archetype

CANDIDATE REGION	HYBRID-CONNECTED	ISOLATED CLOUD-NATIVE / AI	CONNECTED CLOUD-NATIVE / AI	INTERCONNECTED PORTFOLIO
Region A strategic hub region	✓	✓	✓	✓
Region B strategic hub region	✓	✓	✓	◷
Region C in-country, new	✓	✓	⚑	◷
Region D AI model and SKU	✗	✓	⚑	✗
Region E other geography	✗	✗	✗	✗

✓ Qualified ⚑ Conditional ◷ Candidate ✗ Excluded

FIGURE 12 **Illustrative regional qualification by archetype.** Outcomes are dated planning inputs, not permanent approval; validate workload requirements at onboarding. Regions and statuses are illustrative.

> The same region can be qualified for one set of requirements, conditional or a candidate for another, and excluded for a third.

DECISION TOOL · REGIONAL QUALIFICATION

Should this region be qualified?

Apply per candidate region and set of defined workload requirements. Each question is one of the four checks. Use the tree to structure the conversation, not as a gate: most paths are short, and a "no" or "not yet" is a valid, low-cost answer that you record and revisit when conditions change. **Questions 1 and 2 need no technical assessment.** Answer them for every candidate region first, so that deeper work is spent only on the regions that pass.

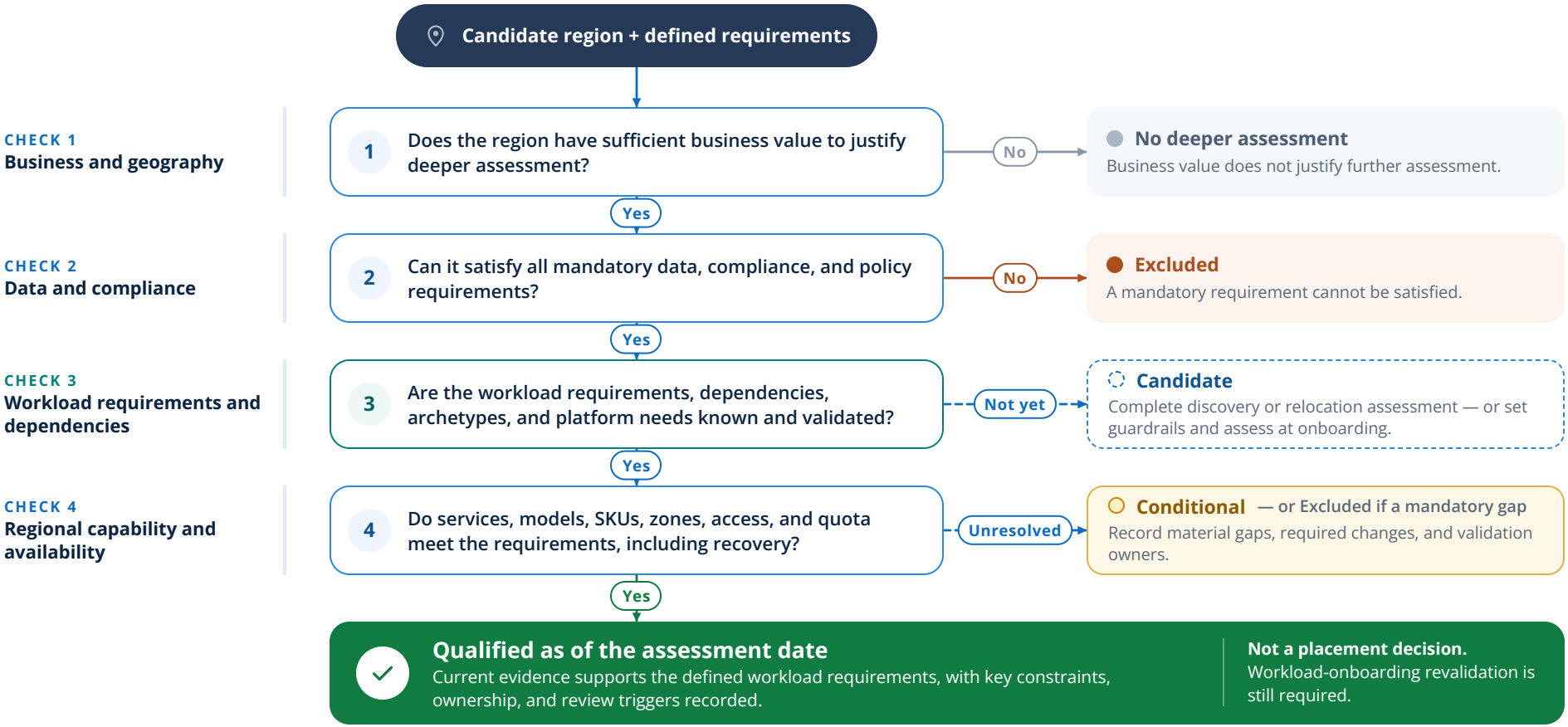


FIGURE 13 **Regional qualification decision tree.** Filters first, then requirements, then technical fit. Candidate and conditional options return to the tree when validation is complete, or conditions change.

Tip. Use the **Multi-region planner** to answer the four checks for each candidate region, compare service availability, prices, and latency, and export to Excel or save a JSON planning record.

For a **qualified** region, the next step is to select its regional design (Part 3).

1 Business and geography

Where do we need Azure presence?

Start with the business reason for considering another region. Evaluate expected markets and users, datacenter and partner locations, planned launches or migrations, anticipated growth, proximity requirements, and geographic concentration risk.

DECISION

Confirm that the region has sufficient business value to justify deeper assessment.

REFERENCE LINKS

- Select Azure regions
- Azure geographies

2 Data and compliance

Are we allowed to operate here?

Determine whether anticipated workloads and their data can operate in the region. Consider data residency and sovereignty, regulatory and industry requirements, company policy, customer commitments, support access, encryption requirements, and restrictions on data movement or operational control.

DECISION

Exclude a region when it cannot satisfy a mandatory data, compliance, or policy requirement.

REFERENCE LINKS

- Data residency in Azure
- Azure compliance documentation
- Reliability and sovereignty in Azure

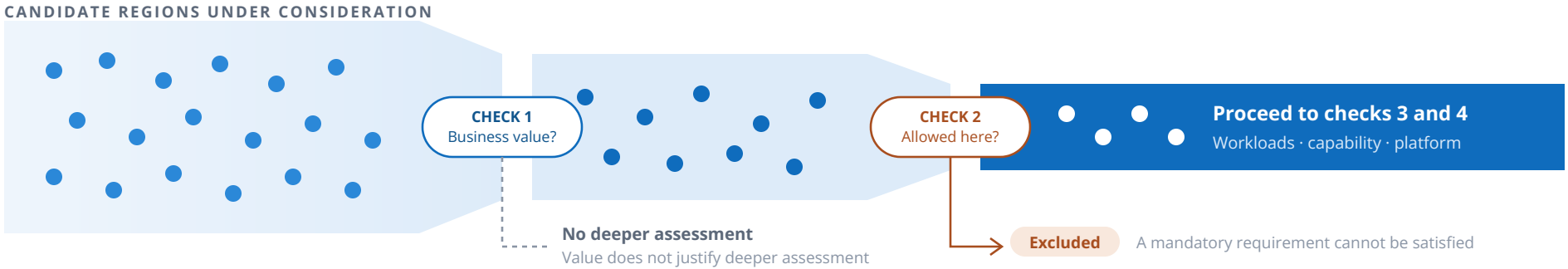


FIGURE 14 **The first two checks filter the portfolio.** Business value decides whether a region merits deeper assessment; mandatory data, compliance, and policy requirements decide whether it can be used at all. Only regions that pass both proceed to workload and capability analysis.

3 Workload requirements and dependencies

What workload requirements and dependencies must the region support?

DECISION

Identify and record the workload-driven platform requirements. They are the input to check 4, which validates what the region can support, and to Part 3, which decides what the platform provides.

Regional qualification should reflect the requirements of the workloads expected to use the region. These come from two sources: existing workloads, including any that would move into the region, and future workloads that are planned but not yet built. This check gathers requirements, but it doesn't decide how they are met, and it doesn't disqualify a region directly.

Review architecture, connectivity, data, shared-service, security, resiliency, scale, and operational requirements and dependencies. Where useful, map the key workloads you expect to move to or deploy in the new region, whether existing or new, to the closest-fitting **Workload Landing Zone archetype**; if a workload has requirements beyond that archetype, record them as explicit additions to it rather than forcing multiple classifications.

For **existing workloads** — in Azure, on-premises, or another cloud — start with the application inventory and architecture. Validate important dependencies with workload owners, first for critical or representative applications and portfolios. Capture application and data dependencies, connectivity, shared services, required Azure services, security and resiliency requirements, and expected scale.

If it would move into the region, from another Azure region or from outside Azure, also identify what must move together, what can stay where it is, and how it behaves while split across locations — latency, data movement, application changes, acceptable disruption, and temporary operation.

For **future workloads**, use known business and technical requirements, predefined workload patterns, intended architecture, expected dependencies, and required services. Where design is still evolving, identify what must influence qualification and defer implementation-specific decisions to onboarding.

SCOPE BOUNDARY

This check doesn't approve relocation or define migration execution; relocation readiness, sequencing, recovery design, and multi-region operation remain workload-specific.

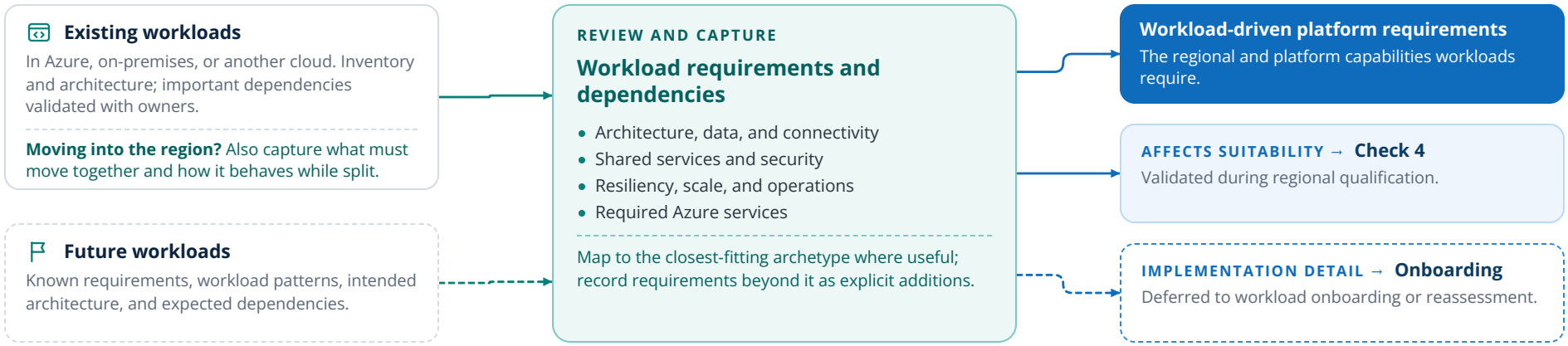


FIGURE 15 **From workloads to platform requirements.** Requirements that could materially affect regional suitability are validated in check 4; implementation-specific decisions can wait for onboarding.

CHECK 3 · CONTINUED

Platform and connectivity requirements

Workload requirements also define what the platform must provide in the region. Capture them in three areas.

Platform enablement

WHAT TO IDENTIFY

The shared services and operational capabilities the workloads depend on, such as identity, security controls, monitoring, backup, and key management, and which of them must be close to the workloads.

DECIDED IN PART 3

Place shared services

Resilient hybrid connectivity

WHAT TO IDENTIFY

The datacenters and users the workloads must reach, and the latency, bandwidth, routing, and resilience they need on that path.

DECIDED IN PART 3

Confirm hybrid connectivity

Regional platform connectivity

WHAT TO IDENTIFY

How the workloads must reach geographic hubs, shared platform services, application dependencies, other Azure regions, and hybrid environments.

DECIDED IN PART 3

Select the connectivity profile

> This check gathers requirements. It doesn't decide how they are met, and it doesn't disqualify a region directly.

REFERENCE LINKS

› Evaluate a cloud workload for relocation

› Relocate cloud workloads

4 Regional capability and availability

Which regions can support the identified workload requirements?

Evaluate each candidate region against the workload requirements identified in check 3. Consider required Azure services, deployment models and SKUs, availability-zone support, regional service dependencies, latency, pricing, expected scale, regional access, and quota requirements.

Assess **availability-zone support** and **region pairing** separately. Prefer zone-enabled regions when zone resiliency is required, but do not assume that region pairing alone provides workload recovery or that a nonpaired region cannot be used as a recovery destination. Validate service-specific capabilities such as replication, backup and restore, and other regional dependencies against the resiliency requirements identified for the workload.

For requirements that depend on multiple regions, validate the intended regional combination at the service level. This includes built-in paired-region behavior such as Storage GRS and region-of-choice capabilities such as Azure SQL replication. Record material gaps, required changes, and owners for unresolved validation; see Service Guides for Multi Region for detailed design.

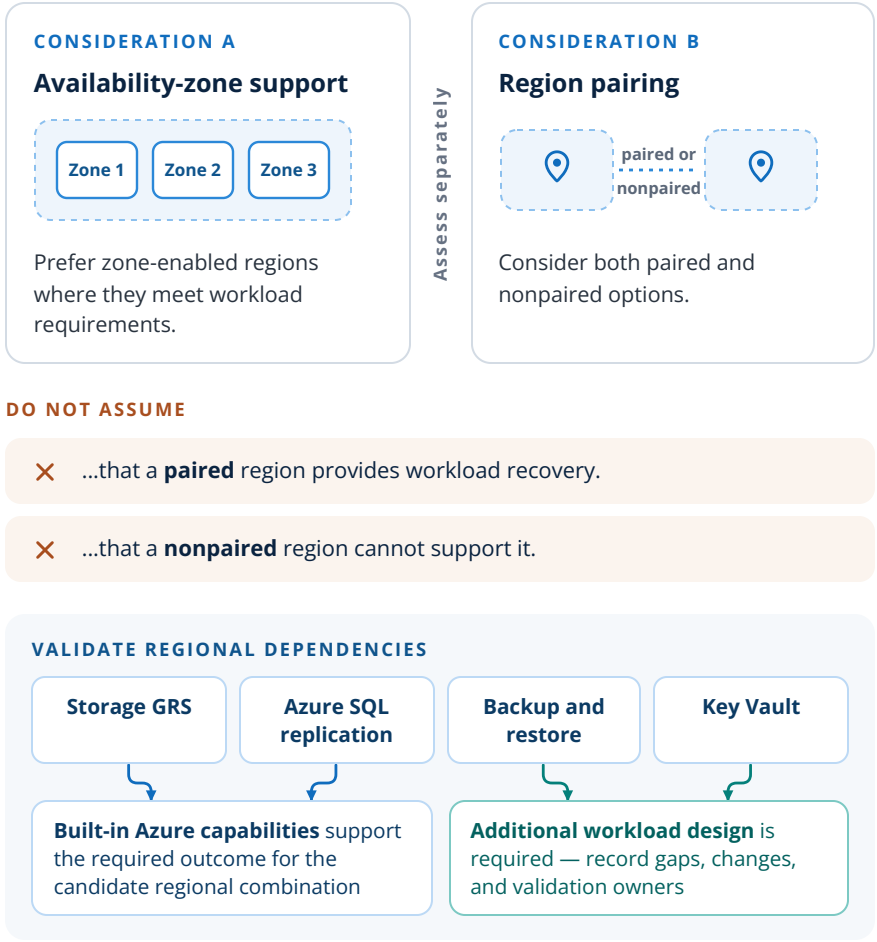


FIGURE 16 **Zones and pairing are independent questions.** Neither guarantees nor rules out workload recovery. See the service guides and Resources.

Continued on the next page: regional access, quota, programmatic evaluation, and the check 4 decision.

CHECK 4 · CONTINUED

Access, quota, and evidence-based evaluation

Validate whether the intended subscription and offer types can deploy in the region, and identify any regional access restrictions or approvals that must be resolved before deployment. Assess required quota based on expected workload scale and identify any increases that may be required.

QUOTA IS NOT CAPACITY

Quota should not be treated as an indication or guarantee of Azure capacity, which remains subject to availability at deployment time.

Evaluate objective regional characteristics programmatically, where possible

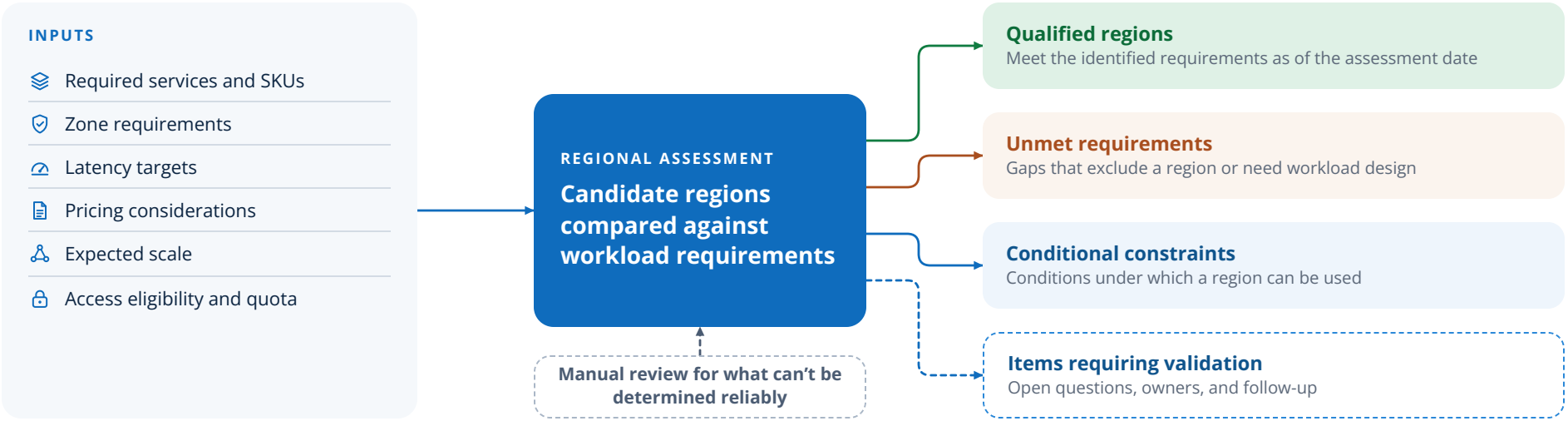


FIGURE 17 **Programmatic regional evaluation.** The resulting assessment should identify qualified regions as of the assessment date, unmet requirements, conditional constraints, and items that require additional validation.

If a requirement that could materially affect regional suitability is unresolved, keep the region **conditional** until it is validated. Evaluate requirements that do not affect regional qualification later during workload onboarding or reassessment.

DECISION

Qualify candidate regions against identified workload requirements, documenting conditions, constraints, and unresolved validations while preserving viable options for future workloads.

REFERENCE LINKS

› Azure region pairs and nonpaired regions › Azure services that support multiple regions › Azure services that support availability zones

› Zonal resources and zone resiliency › Enable zone resiliency for Azure workloads

PLANNING TOOL · STEPS 2 AND 3

Multi-region planner

Compare regional options, record qualification evidence, and select the regional design. The planner supports decisions; it does not qualify a workload automatically or guarantee deployment capacity.



Compare and qualify

Scope current and candidate regions. Compare Services, Latency, and Pricing, then record evidence against the four Checks.

Service, Product, and SKU selectors use an imported availability CSV. Pricing uses a separate Retail Prices API snapshot and service-level comparisons.

The geography view compares region, peering-location, and customer-city latency. Estimated routes do not prove that circuits exist.



Record the design

Regional design records the connectivity profile, shared-service placement, and hybrid connectivity for each selected region.

Hub Connections accepts multiple regions. Existing connectivity can use peering-location suggestions or your own circuit or VPN description.

Enable Advanced Networking Design Tool to explore the inherited design. Tool-only refinements do not modify the authoritative plan; limitations are called out.



Review and export

Review + export collects the evidence and design decisions. Export Excel, print the plan, or save and reopen a JSON planning record.

Visitor-entered planning data stays in the browser; it is not sent to a server. Reference data comes from published snapshots.

Use the full-width step tabs. Open Table of contents to return to the guidance.

REVALIDATE BEFORE ONBOARDING

Availability indicators, snapshot prices, and estimated latency are planning inputs. Confirm current service and SKU support, access, quota, actual capacity, and workload recovery behavior before deployment.

Open the Multi-region planner

03

DECISIONS, NOT DEPLOYMENT

Select the regional design

For each qualified region, make **three choices**, each from a short list. Nothing is built yet.

Selecting the regional design is a set of decisions, not a deployment. For each qualified region, you make three choices, each from a short list. Nothing is built yet: deployment follows later, through the automation you already run.

Each choice follows from the workloads you expect in the region, not from the footprint of an existing hub. Keep one record per region. A region with a complete record is **selected**.

1 Connectivity profile

One of four: Disconnected Spokes, Remote Hub Connected, Minimal Regional Hub, or Full Regional Hub.

Select the connectivity profile **page 30**

2 Shared-service placement

For each shared service the workloads depend on: local, remote, or global.

Place shared services **page 32**

3 Hybrid connectivity

Reuse the existing foundation, change it, or confirm that none is required.

Confirm hybrid connectivity **page 33**

The three choices sit on one path, from on-premises locations to the workloads in the region. They influence one another, but each has different requirements and tradeoffs. Connectivity that is specific to one application isn't decided here. It is designed when the workload is placed (Part 5).



FIGURE 18 **The three decisions on one path** — resilient hybrid connectivity, platform capabilities, and regional platform connectivity. They are related, but each is a separate decision.

Tip. The **Multi-region planner** walks through the three choices for each selected region and produces the regional design record.

PART 3 · REGIONAL DESIGN OVERVIEW

Start from workload requirements

Use the workload requirements identified earlier to determine how workloads should consume regional and centralized platform capabilities. Workload Landing Zone archetypes can help inform the assessment, but the resulting connectivity design should be based on the actual dependencies and requirements of each workload.

- For existing workloads, particularly those with legacy or hybrid dependencies, use validated application, network, datacenter, and shared-service dependencies to determine whether existing geographic hubs can continue to support the workload or whether additional regional capabilities are required.
- For new workloads, derive connectivity needs from the intended architecture and known dependencies, using the Workload Landing Zone archetypes as a planning aid. Reuse existing geographic hubs, shared services, and platform capabilities where they meet those needs; a new region does not automatically require a new platform footprint.

Existing and new workloads follow the same rule: requirements and dependencies determine the regional connectivity profile. Known requirements can shape platform planning in advance; validate workload-specific fit during onboarding and add regional capabilities only when justified.

The regional design record

Keep one record per region. Record the selected profile, the placement of each shared service (and which hub provides the remote ones), the hybrid connectivity decision, the owner, the date, and the triggers that would reopen the decision.

REGION	CONNECTIVITY PROFILE	SHARED SERVICES	HYBRID CONNECTIVITY	OPEN CONDITIONS
Region C	Minimal Regional Hub	Local: selected capabilities. Remote from Region A: remaining shared services.	Reuse: existing foundation where required.	None
Region D	Remote Hub Connected	Remote from a strategic geographic hub.	Reuse: via the hub where required.	Latency test with representative traffic.
Region E	Disconnected Spokes	None required.	Not required.	None

Illustrative example. Regions and choices are examples only.



A region with a complete record is **selected**. Record the open conditions of a conditional region with it, and update the record when a reassessment trigger applies.

PART 3 · CHOICE 1

Select the connectivity profile

What is the most efficient regional platform design that supports the identified requirements?

Determine how workloads in the region will reach geographic hubs, shared platform services, application dependencies, other Azure regions, and hybrid environments. Those connectivity and dependency requirements determine which connectivity profile fits:



Start with the matrix (Figure 9, page 17) to see which connectivity profiles each archetype typically uses. Then apply the selection principles and use the decision tree (page 31) to pick the most efficient profile that meets the requirements.

Profile selection principles

01 Start with the **workload requirements and dependencies** identified during regional qualification, including portfolio-level dependencies. Use Workload Landing Zone archetypes to spot recurring needs, then validate the profile against the actual workloads.

02 Select the **most efficient profile** that satisfies those requirements, not the maximum possible one.

03 Keep capabilities remote when latency, dependency, data, security, resiliency, availability, and operational requirements permit. Add local capabilities only when a measurable requirement justifies the larger regional footprint; adoption growth alone doesn't.

04 Consider direct spoke-to-spoke connectivity where workloads have significant east-west traffic and a hub transit path is unnecessary. Add regional routing, inspection, or shared services when broader requirements justify them.

What each step up costs

How the profiles trade local capability for platform scope

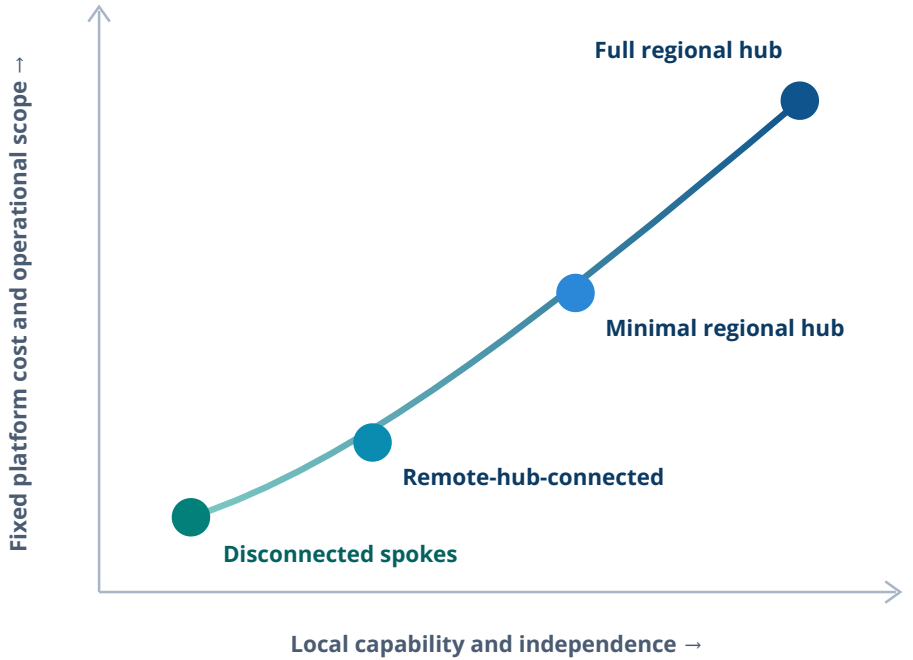


FIGURE 19 **A qualitative trade-off, not a price list.** Greater local capability and independence usually increase fixed platform cost and operational scope, but they can reduce cross-region dependencies, latency exposure, and availability coupling.

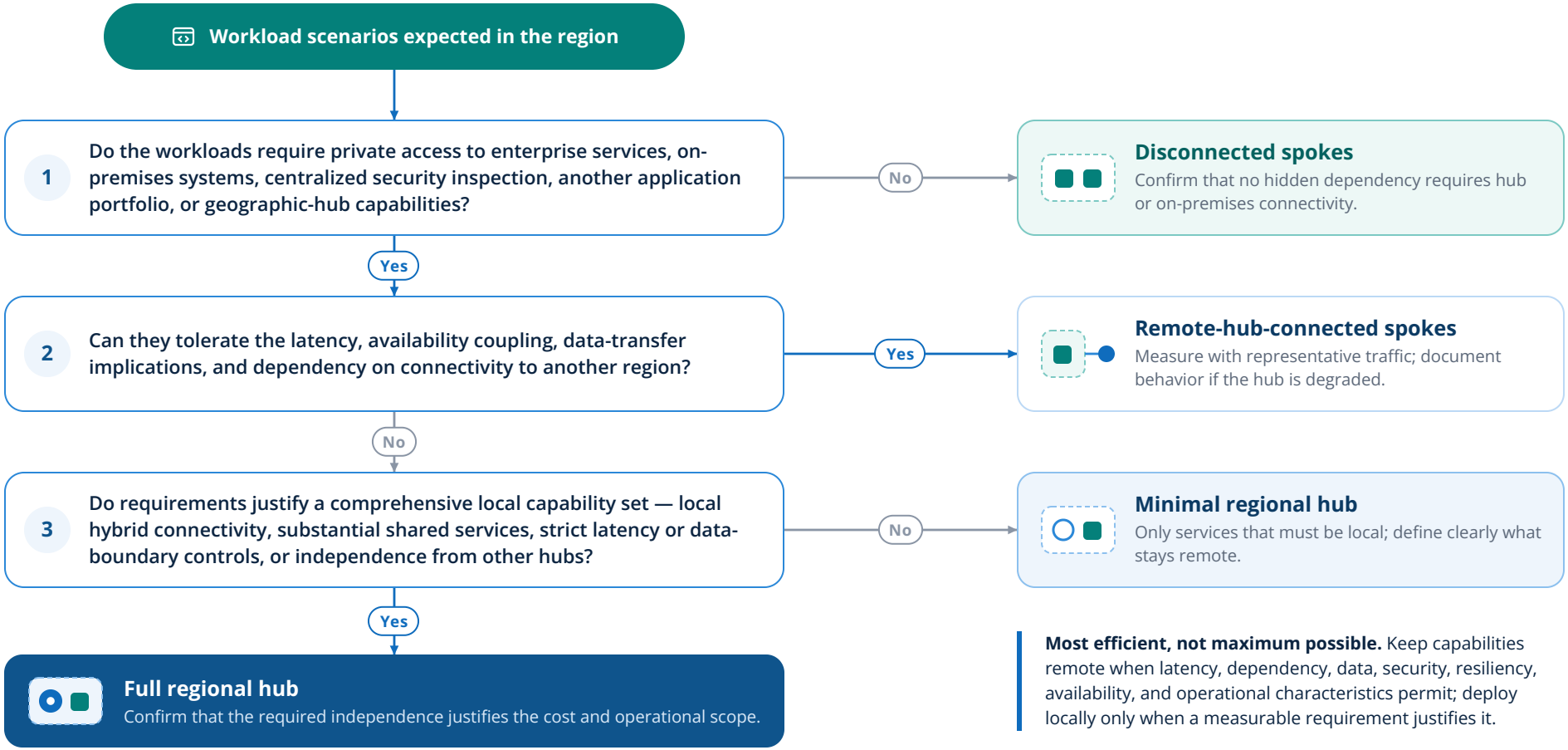
REFERENCE LINKS

- **Network topology and connectivity**
- **Define an Azure network topology**

DECISION TOOL · CONNECTIVITY PROFILE

Which profile should this region provide?

Each question comes from the "fits when" criteria of the profiles. Stop at the first profile that satisfies the requirements identified in check 3. **Two of the four profiles need no local hub**, and only the Full Regional Hub approaches the footprint of an existing geographic hub.



Most efficient, not maximum possible. Keep capabilities remote when latency, dependency, data, security, resiliency, availability, and operational characteristics permit; deploy locally only when a measurable requirement justifies it.

FIGURE 20 **Connectivity profile selection tree.** Questions escalate from dependency, to tolerance of remote consumption, to required independence. The outcome is a platform capability, not a workload design.

DECISION

Select the most efficient regional platform connectivity profile that satisfies the identified requirements and can evolve as regional adoption changes.

Profiles can move in both directions. Check the key considerations before you settle on a profile, and revisit it when its reassessment trigger applies (pages 14–15).

PART 3 · CHOICE 2

Place shared services

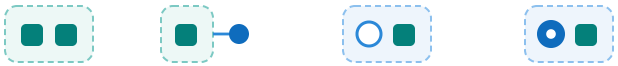
Where is each shared service that the workloads depend on provided?

Determine which shared services and operational capabilities must be provided in the region, consumed from another region, or remain centralized. Each service has one of three placements:

Local	Deployed in the region.
Remote	Consumed from a hub or a shared instance in another region. Record which one.
Global	Not tied to a region. There is nothing to deploy.

Default placement by profile

The table gives a starting point for each profile. The defaults aren't rules: change a cell when a workload requirement calls for it, and record the reason. For a Minimal Regional Hub, each row is a separate choice.



SHARED SERVICE	Disconnected Spokes	Remote Hub Connected	Minimal Regional Hub	Full Regional Hub
Management groups, access control, policy	Global			
Private DNS zones	Global			
DNS resolution across premises Private resolver or forwarders	Not required	Remote	Local or remote	Local
Traffic inspection and internet egress Azure Firewall or network virtual appliance	Not required	Remote	Local or remote	Local
Hybrid connectivity gateways ExpressRoute or VPN	Not required	Remote	Local or remote	Local
Directory services Domain controllers	Not required	Remote	Local or remote	Local
Log workspace	Remote, or local for data residency			
Backup vaults	Local			
Key Vault	Local			

SET BY THE SERVICE, NOT BY THE PROFILE

- **Backup vaults are local in every profile.** Azure Backup works within a region, so you need a vault in each region that holds protected resources.
- **Key Vault is local in every profile.** Give each workload a vault in the region it runs in. See Key Vault across regions in the service guides; design keys separately from secrets and certificates.
- **Private DNS zones are global.** Create them once, in the connectivity subscription, and link virtual networks to them.
- **Start with a single log workspace.** Add a workspace in the region only when data must stay in that geography, or when the charges for sending data to another region are significant.
- **Policy, access control, and management groups aren't tied to a region.** The one change is to allow the new region in any policy assignment that restricts locations.

Also consider operational support, platform resiliency, ownership, and cost. Enable only the regional capabilities required by expected workloads, and document which services remain centralized or are consumed remotely.

REFERENCE LINKS

- Landing zone regions
- Design a Log Analytics workspace architecture
- Private Link and DNS integration at scale
- Hybrid identity in Azure landing zones

DECISION

Define the most efficient platform capability set, where each capability will be provided, who will operate it, and what requirements would trigger future expansion.

PART 3 · CHOICE 3

Confirm hybrid connectivity

Does the operating geography have resilient hybrid connectivity to Azure that satisfies the identified workload requirements?

Enabling an additional Azure region doesn't by itself require another ExpressRoute circuit. Evaluate the existing ExpressRoute, VPN, SD-WAN, or other hybrid connectivity against datacenter and user locations, latency, bandwidth, and routing requirements. Add circuits, peering locations, gateways, or more connectivity paths only where the existing foundation can't provide the required reachability, diversity, or resilience.

THE OUTCOME IS ONE OF THREE

Not required

The region uses Disconnected Spokes, or its workloads need no private path to on-premises locations.

Reuse

The existing circuits, gateways, and peering locations meet the reachability, latency, bandwidth, routing, and resilience that the workloads in the new region need.

Change

A requirement can't be met by the existing foundation: diversity, failure-domain isolation, latency, bandwidth, routing, or resiliency. Record the requirement and the change it justifies.

How a new region reuses what exists

- **A gateway in another region can serve the new region.** Global virtual network peering supports gateway transit, so spokes in the new region can use the gateways of an existing hub.
- **An existing circuit can connect to a gateway in the new region.** An ExpressRoute circuit can connect virtual networks in the same geopolitical region, and across geopolitical regions with the premium add-on. A hub in the new region doesn't need a circuit of its own.
- **Azure Virtual WAN extends in the same way.** A virtual network in the new region can connect to an existing virtual hub in another region, or a new virtual hub in the region can connect to the existing circuits.

DECISION

Confirm the hybrid connectivity foundation and identify any required changes to reachability, resiliency, diversity, scale, or ownership.

When a change is justified

- **The foundation doesn't meet resiliency recommendations today.** Microsoft recommends ExpressRoute circuits from more than one peering location, so that a peering-location outage isn't a single point of failure. A new region is a reason to check this, not a reason to add a circuit for each region.
- **On-premises locations or users are far from the existing peering locations.** Measure latency on the real path before you decide.
- **Bandwidth or gateway capacity is exhausted.** Check the circuit and gateway sizes against the traffic the new region adds.
- **Workloads need a capability that doesn't work across regions.** For example, ExpressRoute FastPath doesn't support global virtual network peering, so workloads that rely on it need a gateway in their own region.
- **The region must keep its on-premises path when another region is unavailable.** That requirement points to a local gateway, and usually to a Full Regional Hub.

REFERENCE LINKS

› Connectivity to Azure › Designing for disaster recovery with ExpressRoute private peering › Use a VPN or ExpressRoute gateway in a different region

› Traditional Azure networking topology › Global transit network architecture and Virtual WAN

04

A TARGETED CHANGE, NOT A NEW LANDING ZONE

Prepare the region

Build only what the selected design calls for. For **two of the four profiles** there is no hub to build.

Preparing a region is a targeted change to the platform you already run, not a new landing zone. You build only what the selected connectivity profile and shared-service placement call for, through your existing landing-zone automation. For two of the four profiles there is no hub to build.

The Cloud Adoption Framework already describes how to **add a new region to an existing landing zone**, including a new hub in the region. That path is the Full Regional Hub. The other three profiles do part of it, for regions whose workloads need less.

What each profile adds

1 Disconnected Spokes



WHAT YOU ADD IN THE REGION

The region in policy and in landing-zone provisioning. Workload spokes with their own internet ingress and egress. No hub, no gateway, no peering.

WHAT YOU REUSE

Governance, identity, policy, monitoring, and automation.

What to deploy **page 36**

2 Remote Hub Connected



WHAT YOU ADD IN THE REGION

The same, plus a cross-region connection from each spoke to an existing hub, with routing and DNS settings. No hub.

WHAT YOU REUSE

An existing geographic hub: its gateways, firewall, DNS, and shared services.

What to deploy **page 37**

3 Minimal Regional Hub



WHAT YOU ADD IN THE REGION

The same, plus a hub in the region with only the components you selected.

WHAT YOU REUSE

The geographic hub, for everything you didn't make local.

What to deploy **page 40**

4 Full Regional Hub



WHAT YOU ADD IN THE REGION

The same, plus a complete hub in the region.

WHAT YOU REUSE

The landing-zone foundation and, where they meet requirements, the existing ExpressRoute circuits.

What to deploy **page 41**

What is global and needs no action

Most of the landing zone isn't tied to a region:

Management groups, subscriptions, and access control

No action. Don't create management groups to model regions.

Policy

Definitions and assignments apply everywhere. Update only the assignments that restrict allowed locations, so that they admit the new region.

Private DNS zones

They are global. Create them once and link virtual networks to them.

Platform subscriptions

Use the existing connectivity and identity subscriptions, with a new resource group for the region.

REFERENCE LINKS

› **Landing zone regions** › **Management groups** › **Subscription vending**

PART 4 · PREPARE THE REGION OVERVIEW

Order of work

The order is the same for every profile. The profile decides how much each item contains.

1. **Allow the region in policy**, for the workload requirements it is qualified for.
2. **Reserve address space** that doesn't overlap the rest of the estate, even for Disconnected Spokes.
3. **Build the network part** that the profile calls for: nothing, connections to an existing hub, or a hub.
4. **Extend the shared services** that must be local, such as DNS forwarding or domain controllers.
5. **Turn on landing-zone provisioning** for the region, so that workload teams receive spokes that match the profile.
6. **Extend operations**: monitoring, security coverage, backup, and runbooks.
7. **Run the platform readiness checklist** (page 43).

WHEN THE REGION IS READY

A region is **ready for workloads** when the required capability set is in place through standard automation, its open conditions are resolved, and every applicable item on the platform readiness checklist is true. Workloads can then onboard without platform exceptions.

Different profiles can coexist across the estate. A region's profile changes only when workload and platform requirements change, not because regions are expected to progress along a maturity path.

Profiles coexisting across one estate

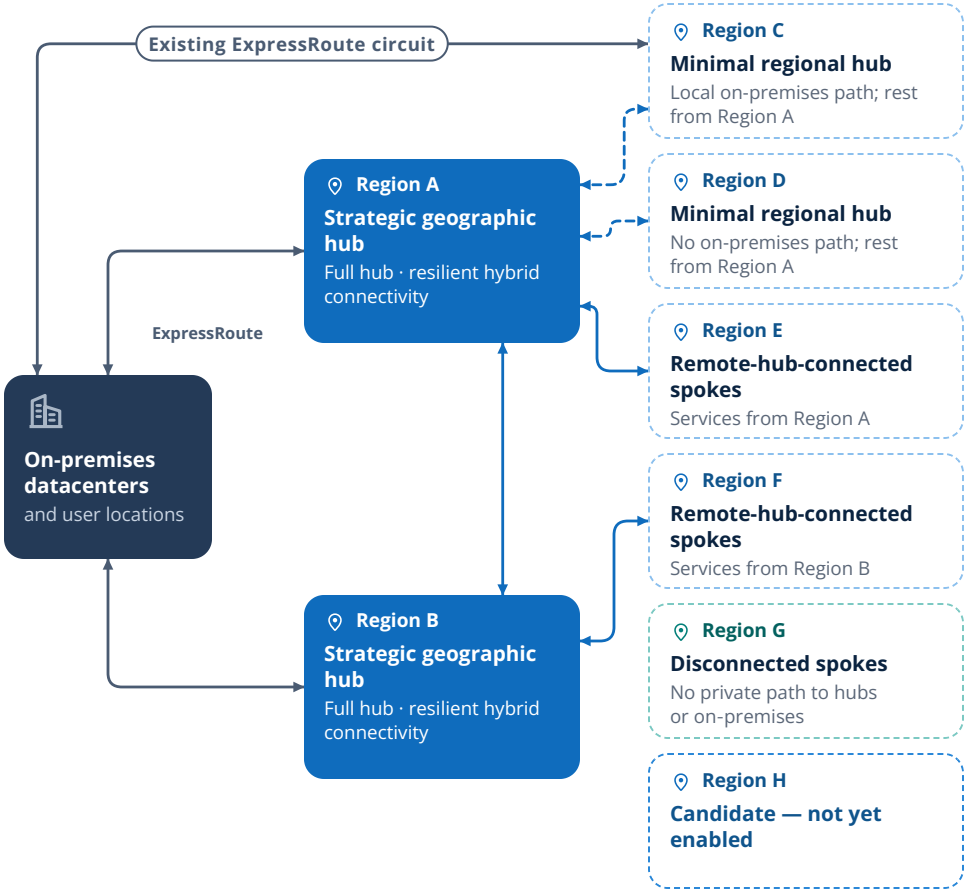


FIGURE 21 **An illustrative regional estate.** Two strategic hubs provide resilient hybrid connectivity; other regions consume what they need. Minimal hubs take selected services from Region A (dashed) — Region C also reaches on-premises by reusing an existing ExpressRoute circuit, Region D does not. Both are valid. Names and placements are examples only.

1

PART 4 · PREPARE THE REGION

Disconnected Spokes



Workload spokes run in the region with no private connectivity to a hub, another Azure region, or an on-premises environment. There is no hub to build. Preparing the region means letting the landing zone you already run accept it.

In the Azure landing zone architecture, these spokes are *online* landing zones: workloads that use direct internet connectivity, or that don't need a virtual network, and that don't connect to the corporate network through a hub.

WHAT YOU DEPLOY

AREA	WHAT TO DO	GUIDANCE
Policy	Add the region to any policy assignment that restricts allowed locations.	**Landing zone regions**
Management groups and access control	Nothing. Place the subscriptions under the management group you use for landing zones without corporate connectivity.	**Management groups**
Landing-zone provisioning	Offer the region in subscription vending, with a spoke virtual network that isn't peered to a hub, or with no virtual network.	**Subscription vending**
Address space	Assign address space that doesn't overlap the rest of the estate.	Platform readiness checklist, page **43**
Internet egress and ingress	Give each spoke explicit outbound connectivity, such as a NAT gateway. Protect inbound HTTP and HTTPS with a web application firewall.	**Plan for inbound and outbound internet connectivity**
Name resolution	Use Azure-provided DNS. For private endpoints, link the spoke virtual network to the private DNS zones it needs.	**Private Link and DNS integration at scale**
Monitoring and security	Send logs to the existing workspace, or to a workspace in the region when data must stay there. Extend security posture and threat detection to the region.	**Design a Log Analytics workspace architecture**
Backup and keys	Create backup vaults and Key Vaults in the region.	Backup and Key Vault (service guides)

WHERE SHARED SERVICES SIT

Global

Management groups, access control, policy, and private DNS zones.

Remote

The log workspace, unless data residency requires one in the region.

Local

Backup vaults, Key Vaults, and each spoke's own egress and ingress.

Not used

A hub firewall, hybrid connectivity gateways, DNS forwarders, and domain controllers.

RECOMMENDATIONS AND CONSIDERATIONS

- **Keep address space unique.** A spoke that isn't peered today can move to a connected profile later only if its address space doesn't overlap.
- **Plan outbound access explicitly.** New virtual networks default to private subnets, without **default outbound access**. A NAT gateway is the recommended method for most workloads that don't need a firewall.
- **Decide who owns private DNS zones.** A disconnected spoke can't use a **DNS resolver in a hub**. Its virtual network needs its own links to the private DNS zones for the private endpoints it uses, whether those zones are the central ones or zones that the workload owns.
- **Plan operator access.** Without a hub there is no shared management path. Decide how operators reach workloads, for example with Azure Bastion in the spoke.
- **Check identity.** Workloads authenticate with Microsoft Entra ID. A workload that needs domain controllers doesn't fit this profile.
- **Connect spokes directly only where needed.** Where applications in the region need east-west communication, evaluate **direct spoke-to-spoke connectivity** without assuming that a regional hub is required. Limit it to spokes of the same environment and workload.
- **Validate the operating procedures.** Validate administration, identity, DNS, secrets, telemetry, data movement, software distribution, private endpoints, incident response, and support procedures.
- **Look for hidden dependencies.** Confirm that no hidden dependency requires connectivity to a geographic hub or on-premises environment.

REASSESS WHEN

Workloads require private access to enterprise services, centralized security inspection, on-premises systems, another application portfolio, or capabilities hosted through a geographic hub.

2

PART 4 · PREPARE THE REGION

Remote Hub Connected



Workload spokes in the region connect across regions to an existing geographic hub and use its gateways, firewall, DNS, and shared services. No hub is built in the region. Preparing the region means connecting its spokes to the hub you already operate.

WHAT YOU DEPLOY

AREA	WHAT TO DO	GUIDANCE
Policy, address space, monitoring, backup, and keys	The same as for Disconnected Spokes (page **36**). Address space must not overlap the hub or the other spokes.	**Landing zone regions**
Landing-zone provisioning	Offer the region in subscription vending, with a spoke virtual network that the platform team connects to the remote hub.	**Subscription vending**
Connection to the hub	Connect each spoke to the existing hub in the other region. See Connect to the existing hub (page **38**).	**Virtual network peering**
Routing and inspection	Apply the same route tables as for spokes in the hub's own region, so that traffic reaches the hub firewall.	**Traditional Azure networking topology**
Name resolution	Point the spokes at the DNS resolver or forwarders in the hub. The private DNS zones are global and don't change.	**DNS for on-premises and Azure resources**
Identity	Use the domain controllers in the hub region, if workloads need them.	**Hybrid identity in Azure landing zones**
Internet ingress	Public application ingress can stay in the region, with a web application firewall close to the workload.	**Plan for inbound and outbound internet connectivity**

WHERE SHARED SERVICES SIT

Global

Management groups, access control, policy, and private DNS zones.

Remote

Gateways, firewall and egress, DNS resolution, domain controllers, and the log workspace, all from the geographic hub.

Local

Backup vaults, Key Vaults, and public application ingress where the workload needs it.

RECOMMENDATIONS AND CONSIDERATIONS

- **Accept the dependency knowingly.** Microsoft's **hub-spoke architecture guidance** recommends at least one hub per region, with only same-region spokes connected to it, so that a failure in one region's hub doesn't affect other regions. This profile trades that isolation for speed and cost. Use it for workloads that can tolerate the dependency, and document how they behave if the geographic hub or cross-region connectivity is degraded or unavailable.
- **Measure before you commit.** Measure latency and bandwidth using representative traffic.
- **Count the cross-region cost.** Global virtual network peering carries traffic charges. Evaluate cross-region data-transfer cost, routing and transit behavior, and inspection paths.
- **Check the hub's capacity.** Evaluate geographic hub scale: gateway and firewall throughput, peering limits, and dependency criticality.
- **Agree who operates the path.** Agree operational ownership for the connection, routes, and DNS settings between the team that runs the hub and the teams in the new region.
- **Keep east-west traffic local where it matters.** For spokes in the region that exchange a lot of traffic, direct connectivity between them avoids a round trip through the remote hub. Limit it to spokes of the same environment and workload.

REASSESS WHEN

Cross-region traffic, latency, east-west communication, dependency criticality, regulatory requirements, scale, or availability coupling justify deploying selected connectivity or shared-service capabilities locally.

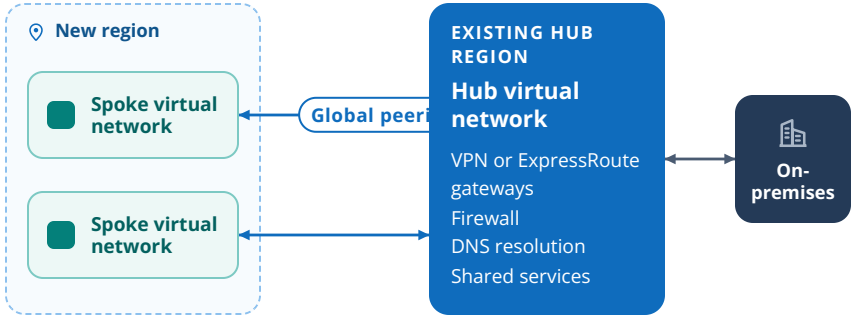
2 PART 4 · PREPARE THE REGION

Remote Hub Connected continued

Connect to the existing hub

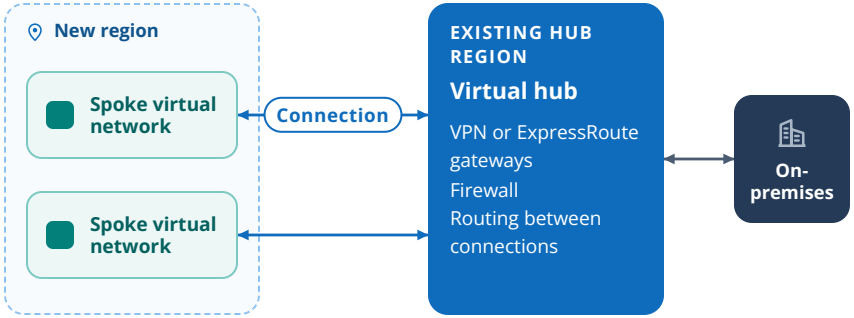
How you connect depends on the topology of the hub you already run.

Hub-and-spoke virtual networks



Create global virtual network peering between each spoke and the hub virtual network. Turn on **gateway transit** when the spokes use the hub's VPN or ExpressRoute gateways. Gateway transit works over global peering.

Azure Virtual WAN



Connect each spoke virtual network to the existing virtual hub. Virtual WAN supports **connecting a virtual network in one region to a virtual hub in another**, and describes it as a good option when the footprint is mostly in one region with a few remote spokes.

FIGURE 22 **Two ways to reach the hub you already run.** Spokes in the new region use the gateways, firewall, and shared services of a hub in another region. No hub is built in the new region.

CHECK THESE LIMITS BEFORE YOU RELY ON THE CONNECTION

- A spoke that uses remote gateways can't have its own gateway, and it can use the gateways of only one peering.
- ExpressRoute FastPath doesn't support global virtual network peering.
- Resources can't reach the front end of a basic load balancer across global peering.
- Peering isn't transitive. Spokes that peer to the same hub don't reach each other through peering alone; they route through the hub firewall or connect directly.

PART 4 · PREPARE THE REGION

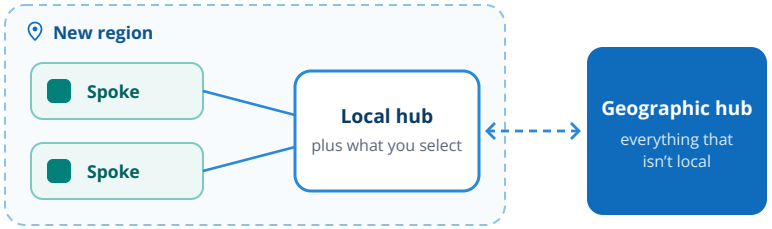
3 Minimal Regional Hub

A Minimal Regional Hub is a hub in the region that holds only what its workloads need locally. Everything else comes from a geographic hub. It isn't a fixed design. Which components are local is a choice for each region. The fewer components you make local, the faster the region is ready and the less it costs to run.

THE BASE

Every Minimal Regional Hub has the same base:

- A hub in the region, in the connectivity subscription you already use, with a new resource group for the region. The hub is a hub virtual network, or a virtual hub in your existing Azure Virtual WAN.
- A connection from that hub to the geographic hub.
- Connections from the spokes in the region to the local hub.
- The routing that sends traffic to the right hub.



The base of a Minimal Regional Hub. Spokes connect to the local hub, which connects to the geographic hub.

EXAMPLE COMPOSITIONS

These are examples, not designs to copy. Each follows from a workload reason.

Local on-premises path. A hub with an ExpressRoute gateway connected to an existing circuit. Inspection, DNS, and identity stay in the geographic hub. The reason is latency-sensitive traffic to a nearby datacenter.

Local inspection and egress. A hub with a firewall and DNS forwarding, and no on-premises path of its own. The reason is heavy traffic between spokes in the region, or internet egress that must leave from the region.

Local shared services only. A hub virtual network that hosts DNS forwarding and connects to domain controllers in the region, with no firewall and no gateway. The reason is applications that must keep authenticating and resolving names when the link to the other region is down.

OPTIONAL COMPONENTS

Add a component only when a workload requirement calls for it in the region.

COMPONENT	MAKE IT LOCAL WHEN	KEEP IT REMOTE WHEN	GUIDANCE
ExpressRoute or VPN gateway	Workloads need an on-premises path from the region for latency, bandwidth, or independence. The gateway can connect to an existing circuit.	On-premises traffic is light or tolerates the path through the geographic hub.	Confirm hybrid connectivity, page **33**
Azure Firewall or network virtual appliance	Traffic between spokes in the region is heavy, internet egress must leave from the region, or data must be inspected inside a boundary.	Inspection in the geographic hub meets latency, cost, and data requirements.	**Plan for inbound and outbound internet connectivity**
DNS private resolver or forwarders	Name resolution for the region must not depend on the link to another region.	Spokes can use the resolver in the geographic hub.	**DNS for on-premises and Azure resources**
Domain controllers	Applications authenticate against them often, or must keep working when the other region is unavailable.	Authentication traffic is light and tolerates the dependency.	**Hybrid identity in Azure landing zones**
Other shared services	A requirement for latency, residency, or independence names the service.	No requirement names it.	Place shared services, page **32**

3

PART 4 · PREPARE THE REGION

Minimal Regional Hub continued



WHAT YOU DEPLOY

AREA	WHAT TO DO	GUIDANCE
Policy, address space, landing-zone provisioning, monitoring, backup, and keys	The same as for Disconnected Spokes (page **36**), with spokes that connect to the local hub.	**Landing zone regions**
Hub (the base)	Create the hub virtual network or virtual hub, connect it to the geographic hub, connect the spokes, and configure routing.	**Traditional Azure networking topology** or **Virtual WAN network topology**
Optional components	Deploy the components that must be local, in the same order as for a Full Regional Hub (page **41**): inspection, gateways, name resolution, then identity.	**Landing zone regions**

HUB DESIGN

The same choices apply as for a full hub, on a smaller scale:

- **Topology.** **Hub-and-spoke virtual networks** or Azure Virtual WAN. Use the topology the rest of the estate uses.
- **Inspection.** Azure Firewall or a third-party network virtual appliance, if the hub has inspection at all.
- **Transit.** Integrated in the hub, or part of the organization's SD-WAN.

For the options in full, see Hub design options (page **42**).

REASSESS WHEN

Workload growth introduces additional local connectivity or shared-service requirements, multiple interconnected application portfolios require broader regional capabilities, stricter independence requirements emerge, significant hybrid connectivity is required locally, or other requirements justify expanding the regional platform footprint.

RECOMMENDATIONS AND CONSIDERATIONS

- **Write down the split.** Clearly define which capabilities operate locally and which remain remote. Avoid ambiguous routing, overlapping responsibilities, and unnecessary duplication of platform controls.
- **Plan the on-premises path when the hub has no gateway.** Virtual network peering isn't transitive, so spokes don't reach the geographic hub's gateways through the local hub automatically. Connect a gateway in the local hub to an existing circuit, peer the spokes that need the path to the geographic hub as well and **use its gateways**, or use Virtual WAN, which provides transit between hubs. Routing through network virtual appliances in both hubs is possible, but the region's address space must then be advertised to on-premises.
- **Reuse circuits before you add them.** When connectivity to on-premises environments is required, determine whether existing ExpressRoute circuits and gateway topology can provide the required reachability before introducing additional circuits; do not assume that each Azure region requires a dedicated ExpressRoute circuit.
- **Test both failure directions.** Validate workload and operational behavior when either local capabilities or remote dependencies are unavailable.
- **Watch for inspection being bypassed.** When hubs in different regions share ExpressRoute circuits, spokes in different regions can learn routes to each other and bypass the firewalls. Plan route tables so that traffic between regions is inspected where it must be.
- **Know which profile you have.** The hub is minimal as long as the region still depends on a geographic hub for something. When it can keep operating without the other hubs, it is a Full Regional Hub, and its cost and operating scope should be planned as one.

PART 4 · PREPARE THE REGION

4 Full Regional Hub



The region gets a complete hub: local connectivity, inspection, DNS, identity, and the other shared services its workloads need. It can keep operating when other hubs are unavailable. This is the path that the Cloud Adoption Framework describes for adding a region to an existing landing zone, so most of the work is documented there. A Full Regional Hub uses the same operating model and control baselines as your other hubs. It doesn't have to be an identical copy.

WHAT YOU DEPLOY

Start with what every profile needs: the region in policy, address space, landing-zone provisioning, monitoring, backup, and keys (page **36**). Then build the hub in this order.

STEP	HUB-AND-SPOKE VIRTUAL NETWORKS	AZURE VIRTUAL WAN
1. Hub	In the existing connectivity subscription, create a resource group and a hub virtual network in the region.	In the existing Virtual WAN, create a virtual hub in the region.
2. Inspection	Deploy Azure Firewall or network virtual appliances in the hub.	Deploy Azure Firewall or a supported appliance in the virtual hub, with routing intent and routing policies.
3. Hybrid connectivity	Deploy VPN or ExpressRoute gateways and connect them, reusing existing circuits where they meet requirements.	Deploy the gateways in the virtual hub and connect them.
4. Connection to other hubs	Peer the new hub virtual network with the other hub virtual networks.	Automatic between hubs of the same Virtual WAN.
5. Routing	Configure Azure Route Server or user-defined routes.	Configure any static routes the virtual hub needs.
6. Name resolution	Deploy **DNS forwarders** for the region and link them to the private DNS zones.	Deploy them in a spoke connected to the virtual hub.
7. Identity	In the identity subscription, create a virtual network in the region, peer it to the hub, and deploy domain controllers if workloads need them.	The same.
8. Spokes	Peer workload spokes in the region to the new hub.	Connect them to the new virtual hub.

For the detail of each step, follow **Add a new region to an existing landing zone**. The network steps are described in **Traditional Azure networking topology** and **Virtual WAN network topology**.

CONNECT THE HUB TO THE REST OF THE ESTATE

Two regions.
Connect the hub virtual networks with global virtual network peering.

More than two regions.
Connect the hub virtual networks in each region to the same ExpressRoute circuits, or use Azure Virtual WAN for managed transit between regions.

A few workloads across regions.
Connect the landing-zone virtual networks that need each other directly with global peering, instead of routing all traffic through the hubs.

WHERE SHARED SERVICES SIT

Global

Management groups, access control, policy, and private DNS zones.

Local

Gateways, firewall and egress, DNS resolution, domain controllers, backup vaults, and Key Vaults.

Local or remote

The log workspace. Start with the existing workspace unless data must stay in the region.

PART 4 · PREPARE THE REGION

4 Full Regional Hub continued

RECOMMENDATIONS AND CONSIDERATIONS

- **Confirm the hub is justified.** Confirm that the required level of independence justifies the additional cost and operational scope.
- **Reuse circuits before you add them.** Where on-premises connectivity is required, determine whether existing ExpressRoute circuits can provide the required reachability and resiliency before introducing additional connectivity.
- **Build hybrid connectivity to the resiliency recommendations.** Use circuits from more than one peering location and zone-redundant gateways where the region supports them. See **Connectivity to Azure**.
- **Remove the dependencies you built the hub to avoid.** A full hub that still resolves names, authenticates, or sends egress through another region isn't independent. Test the region with the other hubs unavailable.
- **Plan inspection between regions.** When hubs share ExpressRoute circuits, spokes in different regions can learn routes to each other and bypass the firewalls. Use route tables to send that traffic through the firewall where it must be inspected.
- **Operate it like the other hubs.** Validate automation, observability, support ownership, security operations, configuration consistency, dependency behavior, and lifecycle management.

REASSESS WHEN

Workload demand or requirements decrease, services can be safely centralized, dependencies change, or the full regional footprint no longer provides sufficient value relative to its cost and operational complexity.

HUB DESIGN OPTIONS

Profiles describe capability, not a specific network build. The same profile can be implemented in different ways:

Hub implementations vary between organizations

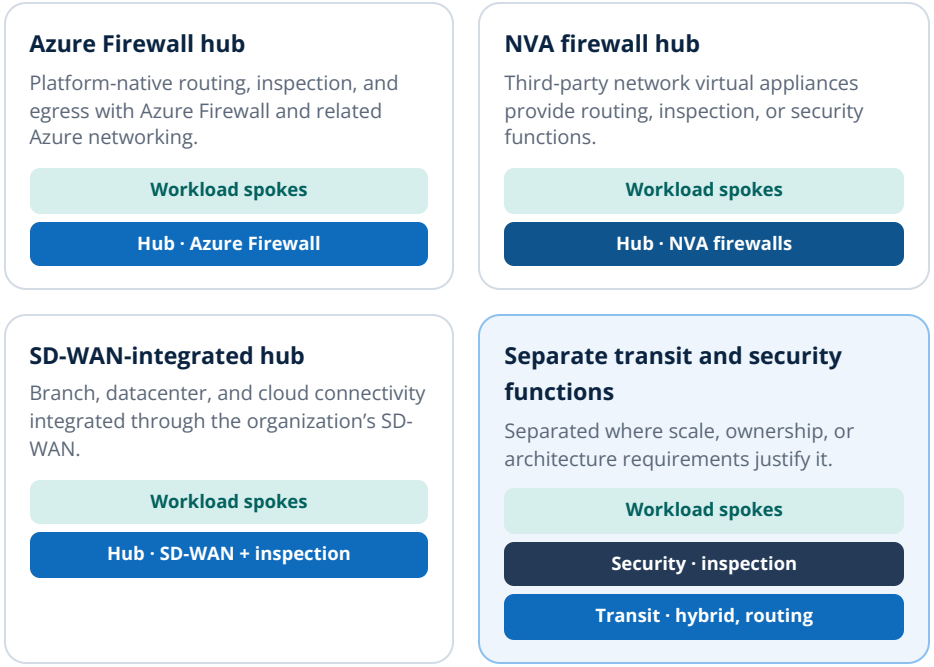


FIGURE 23 **Hub implementations vary between organizations.** Any of these designs can sit behind a Minimal or Full Regional Hub.

TOPOLOGY NOTES

The profiles can be implemented using **hub-and-spoke virtual networks**, Azure Virtual WAN, or other approved Azure networking patterns. Topology determines how the required capability is implemented; workload and platform requirements determine which profile is appropriate.

PART 4 · BEFORE WORKLOADS ARRIVE

Platform readiness checklist for a new region

These are the platform items that most often stall a new region after it has been qualified. Most are inexpensive to plan early and expensive to retrofit. Get these right once, and each later region becomes mostly configuration instead of a new project. Apply only those the region's connectivity profile and workload requirements call for.

Network foundation

- ☐ **IP address planning**
  Reserve non-overlapping address space for future regions and profiles.
- ☐ **Hybrid connectivity**
  Existing circuits are reused, or paths are added (page 33).
- ☐ **Private DNS across regions**
  Private endpoints and hybrid names resolve wherever the workload runs.
- ☐ **Firewall and partner allow-lists**
  On-premises firewalls and third-party allow-lists include the new region's address ranges.
- ☐ **Routing and inspection paths**
  Traffic reaches hubs and firewalls without asymmetric paths.
- ☐ **Identity services**
  The directory and identity services that workloads depend on are reachable from the region.

Governance and automation

- ☐ **Region settings in policy**
  Allowed-locations and region parameters admit the new region only for its qualified workload requirements.
- ☐ **Region-aware infrastructure as code**
  Modules, naming, and tagging are parameterized by region, with no hard-coded locations.
- ☐ **Landing-zone provisioning**
  Subscription or landing-zone vending can place workloads in the region.
- ☐ **Regional access**
  Subscriptions have access to the region where access is restricted.

Operations and data

- ☐ **Where logs and telemetry live**
  Workspace location and residency for monitoring and security data are decided.
- ☐ **Backup placement**
  Vaults and valid restore targets exist for the region. See Backup and restore in the service guides.
- ☐ **Runbooks and support**
  Incident, on-call, and support procedures include the region.

Capacity and security

- ☐ **Quota and specific capacity**
  Quota for the required SKUs and models is requested ahead of onboarding; capacity reservations are used where justified.
- ☐ **Key Vault placement**
  Workload vaults exist per region rather than as a cross-region dependency. See Key Vault in the service guides.
- ☐ **Security coverage**
  Security posture, threat detection, and privileged access paths extend to the region.

☐ Common blockers: plan early, during qualification ☐ Confirm as the capability set is put in place

Private DNS, routing and inspection paths, and hybrid connectivity apply only to connected profiles; everything else applies to every profile, including Disconnected Spokes.

FIGURE 24 **Platform readiness checklist.** Hub and shared-service implementations differ; the checklist names what must be true, not how a specific hub is built.

✓ **Ready for workloads** means the region is qualified for its workload requirements, the required capability set is in place through standard automation, and workloads can onboard without platform exceptions.

FROM OPTIONS TO PLACEMENT

05 Place workloads using qualified options

Previous regional qualification is a planning input; **decision-time validation is authoritative.**

Place a workload only after its requirements are understood well enough to make a decision. Regional qualification narrows the choices; it **does not permanently approve** a region for every application. Placement ends in one of these outcomes: place the workload in a qualified region, keep it where it is, relocate it, extend it to another region for recovery or active use, or conclude that no qualified option meets its requirements yet.

Apply the same eight steps to each workload or related application portfolio. The workload team works within the qualified options; additional platform capabilities, where needed, are a separate governed decision. Findings flow back into regional qualification.

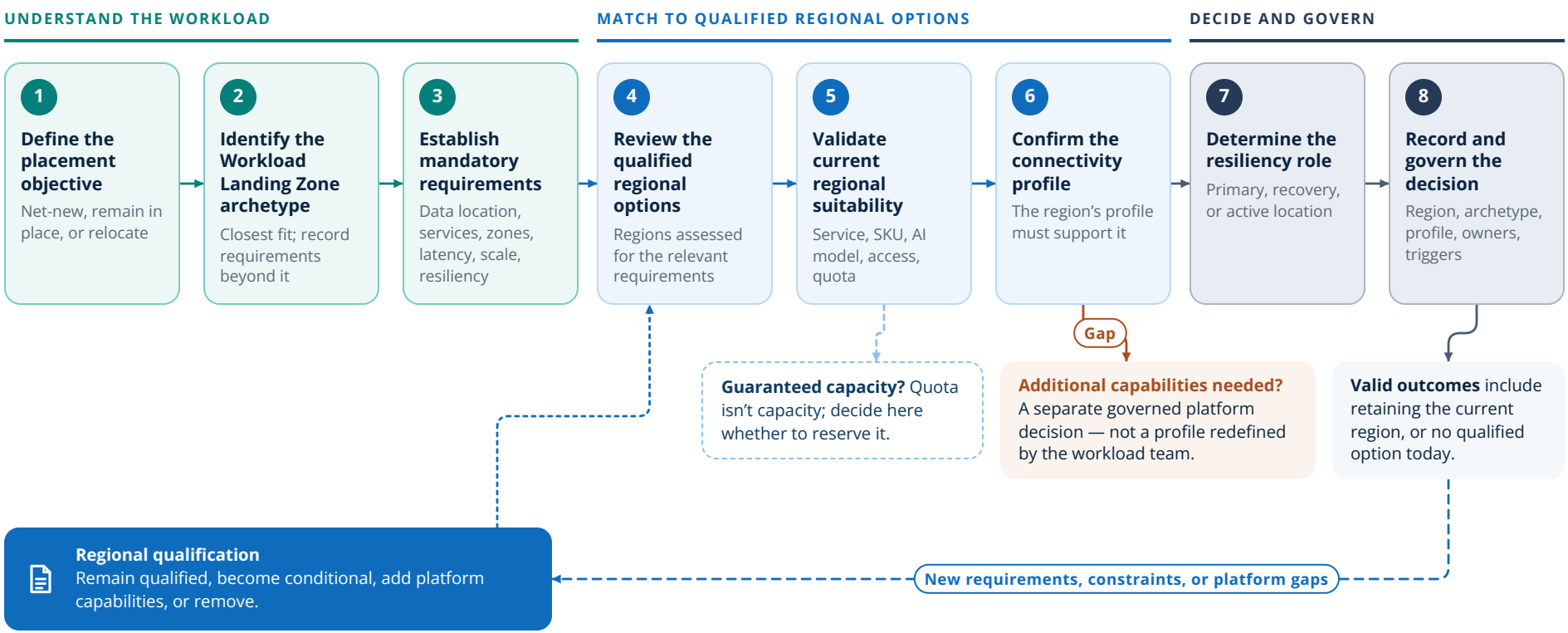


FIGURE 25 **Workload placement using qualified options.** Teal steps describe the workload, blue steps match it to regional options, and slate steps decide and govern. Onboarding findings return to regional qualification, closing the loop between platform readiness and workload adoption.

THE EIGHT PLACEMENT STEPS

01 Define the placement objective

Is the workload new, staying where it is, or moving or expanding to another region?

02 Identify the Workload Landing Zone archetype

Assess the workload's connectivity, shared-service, data, security, operational, and dependency characteristics and pick the closest-fitting archetype. Record any material requirements it doesn't cover as explicit additions to it.

03 Establish mandatory requirements

Confirm data-location restrictions, required Azure services and features, availability-zone requirements, latency thresholds, expected scale, connectivity, resiliency requirements, and critical dependencies.

04 Review the qualified regional options

Consider regions previously assessed against the relevant workload requirements. Review their supported capabilities, connectivity profiles, readiness, known constraints, when each was last assessed, and any outstanding validations.

05 Validate current regional suitability

Qualification was an earlier planning assessment, so **re-check the target region before onboarding**, whether the workload is new or moving. Confirm that the services, SKUs, AI models, regional access, quota, and dependencies it needs are available now. Quota allows deployment but doesn't guarantee capacity; if the workload needs guaranteed capacity, decide here whether to reserve it.

06 Confirm the region's connectivity profile supports the workload

Check that the region's connectivity profile satisfies the workload's requirements. If additional platform capabilities are required, treat their introduction as a separate governed platform decision rather than allowing the workload team to redefine the regional profile independently.

07 Determine the workload's resiliency role

Define whether the region supports the workload as its primary location, recovery location, or part of a multi-region active deployment. Apply workload-specific availability, RTO, RPO, replication, data-consistency, and failover requirements. See the service guides.

08 Record and govern the decision

Document the selected region, archetype, connectivity profile, placement and resiliency outcomes, assumptions, constraints, owners, and revalidation triggers.



FIGURE 26 **Qualification versus placement.** The platform qualifies regions against defined requirements; each workload decides where and how it actually runs. Each informs the other.

Feed findings back into qualification

Workload onboarding can reveal new regional requirements, constraints, or platform gaps. Use these findings to update regional qualification and determine whether an option should remain qualified, become conditional, require additional platform capabilities, or be removed from consideration for the affected workload requirements.

PRINCIPLE

Regional adaptability does not mean arbitrary placement.

SERVICE GUIDES

Service Guides for Multi Region

A multi-region platform creates qualified regional options. Whether a specific workload uses more than one of those regions, and how, is a workload design decision.

This section isn't workload design guidance. The Azure Well-Architected Framework covers how to design a multi-region workload.

This section covers the regional and service behaviors a platform team should understand when it qualifies regional combinations: region pairing is one input into resiliency design, not the definition of it, and cross-region behavior can differ significantly by service and regional combination.

Where workload design guidance lives

AZURE WELL-ARCHITECTED FRAMEWORK

Using availability zones and regions

How to choose a single-region, zonal, or multi-region design for a workload.

AZURE WELL-ARCHITECTED FRAMEWORK

Architecture strategies for designing for redundancy

Multi-region deployment models, including active-active and active-passive.

AZURE WELL-ARCHITECTED FRAMEWORK

Design methodology for mission-critical workloads

The full multi-region active design for the most demanding workloads.

AZURE RELIABILITY

Multi-region solutions in nonpaired regions

Service-by-service guidance for regions without a pair.

AZURE RELIABILITY

Azure reliability service guides

Regional behavior for each Azure service.

In this section

The pages that follow cover the dependencies that most often decide whether a regional combination works as built.

Storage

Paired-region and region-of-choice behavior for Azure Storage replication.

Backup and restore

Cross Region Restore, Azure Site Recovery, and region-of-choice recovery.

Key Vault

Product selection, secrets and certificates, five key patterns, and a decision tree.

Private DNS zones

Global and regional zone designs for resources with private endpoints in multiple regions, including failover and policy considerations.

For related background, see Paired and nonpaired regions and Resiliency scenarios in Resources.

SERVICE GUIDES · STORAGE

Storage across regions

Storage replication illustrates why regional qualification cannot stop at “the service exists in both regions.”

Paired-region options

- Geo-redundant storage options such as GRS and GZRS replicate data asynchronously to the service-defined secondary region.
- Built-in geo-redundancy and failover behavior must be evaluated separately from workload-level recoverability.

Region-of-choice options

- Object replication can asynchronously replicate **block blobs** between selected storage accounts and regions.
- It requires change feed on the source and blob versioning on source and destination accounts.
- Object replication does not support append blobs, page blobs, or accounts with hierarchical namespace enabled.
- A source account can currently replicate to no more than two destination accounts.

VALIDATE THE COMPLETE RECOVERY PATH

Replicated data does not by itself make the workload recoverable. Validate the application recovery path end to end.

Azure Storage redundancy · Object replication for block blobs

SERVICE GUIDES · BACKUP AND RESTORE

Backup and restore across regions

Backup and restore capabilities differ by workload and region combination, so confirm them for each intended recovery region.

Paired-region capabilities

- Azure Backup Cross Region Restore uses the Azure paired region for supported data sources when the required vault replication and configuration are enabled.
- Support and behavior differ by backup workload and configuration.

Region-of-choice capabilities

- Determine which backup technologies and restore targets support the intended recovery region.
- Azure Site Recovery supports Azure VM disaster recovery between Azure regions, subject to its support matrix and regional restrictions.
- Restricted-access regions can require explicit access approval.

PROVE THE RECOVERY OBJECTIVES

Demonstrate RTO and RPO through recovery testing, not configuration alone.

Restore Azure VMs, including Cross Region Restore · Azure Site Recovery support matrix

SERVICE GUIDES · KEY VAULT

Azure Key Vault across regions

Key Vault is often a runtime dependency. Pick the product first, then the pattern. Each product handles regions differently.

	Key Vault Standard	Key Vault Premium	Managed HSM	Managed HSM + EKM
PROTECTION	Software, FIPS 140 Level 1	HSM-backed keys, FIPS 140-3 Level 3	Single-tenant HSM, FIPS 140-3 Level 3	Key stays in your own HSM outside Microsoft
SLA	99.99%	99.99%	99.9% in one region; 99.99% when replicated across regions	No end-to-end SLA; the Managed HSM SLA covers only the service
HOLDS	Asymmetric keys, secrets, certificates	Asymmetric and symmetric keys (preview), secrets, certificates	Asymmetric and symmetric keys	Asymmetric and symmetric keys
ZONES	Automatic zone redundancy in zone regions	Automatic zone redundancy in zone regions	Not zone-based; partitions sit on separate servers in different racks	Depends on your HSM
REPLICATION	To the paired region only (same geography); none without a pair	To the paired region only (same geography); none without a pair	Multi-region replication to one extended region, paired or not	None; EKM does not support geo-replication
FAILOVER	Microsoft-managed; read-only after failover	Microsoft-managed; read-only after failover	Both regions stay active; reads and writes continue	No failover target; recovery depends on the failing part

PAIRED-REGION FAILOVER · STANDARD AND PREMIUM

Outage begins
0

Public endpoints fail over
a few minutes

Private endpoints follow
up to +20 minutes

Vault is read-only
until failover ends

Microsoft replicates vault contents to the paired region and runs a tested failover. Some pairs, and all nonpaired regions, do not use this model. For the current list, see Reliability in Azure Key Vault.

REGIONAL DESIGN CONSTRAINTS

3+0 **No paired region**
Qatar Central, Belgium Central, Spain Central

3+1 **Paired, asymmetric**
West US 3 → East US

3+1 **Paired outside the geography**
Brazil South ↔ South Central US

DATA CAN BE SAFE WHILE THE APP IS DOWN

Applications can require keys, secrets, and certificates simply to initialize, so data can be recoverable while the application still cannot start.

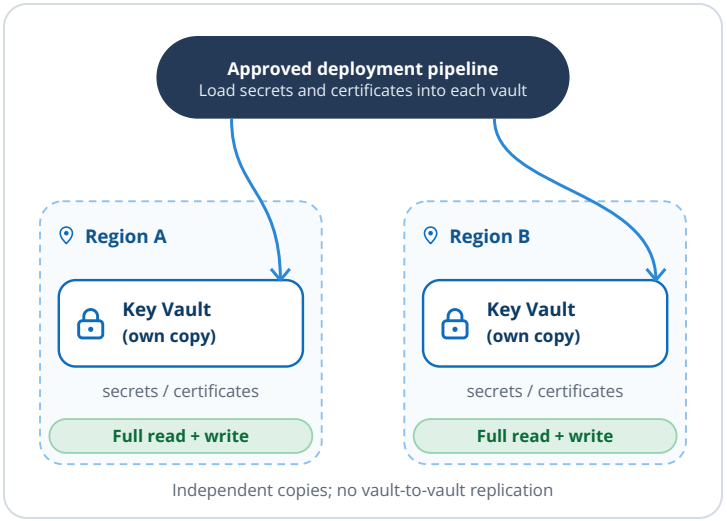
SERVICE GUIDES · KEY VAULT · SECRETS AND CERTIFICATES

Secrets and certificates: two patterns

Secrets and certificates can be loaded into any vault by a pipeline. That makes a vault in each region practical.

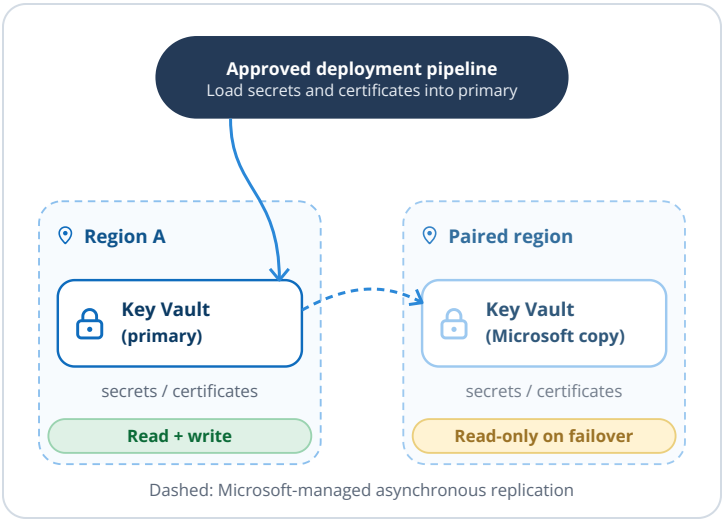
1 Key Vault in each region

CREATED IN	Each vault. CI/CD and IaC load the same secrets and certificates into every vault.
OUTAGE	Each region keeps full read and write on its own vault.
BEST FOR	Mission-critical workloads that need a short recovery time.
WATCH FOR	You must keep the vaults in sync with your pipeline.



2 Key Vault in a paired region

CREATED IN	One vault. Microsoft fails it over to the paired region.
OUTAGE	Read-only until failover ends: you can read, but not create, update, or rotate.
BEST FOR	Less critical workloads that accept read-only access and Microsoft's failover timing.
WATCH FOR	Fails over only to a pair in the same geography that is paired in both directions.



DESIGN GUIDANCE

Treat each region as its own deployment stamp with its own vault. Roll secrets out through a pipeline, one region at a time, so one bad change cannot reach every region. App-level fallback to another region's vault is optional and needs careful design of networking (private endpoints) and identity (managed identities).

Managed HSM and EKM do not hold secrets or certificates. Use Standard or Premium for both.

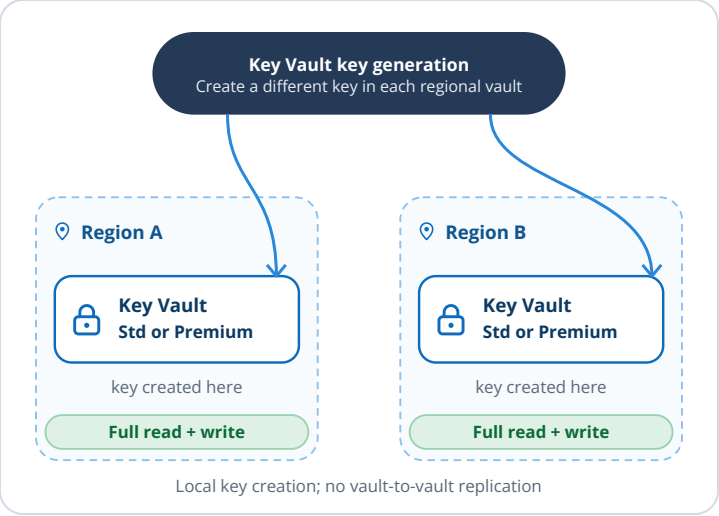
SERVICE GUIDES · KEY VAULT · KEYS (1 OF 2)

Keys: vault-based patterns

Patterns 1 and 2 use Key Vault. Customer-managed keys have extra limits, so design them separately from secrets.

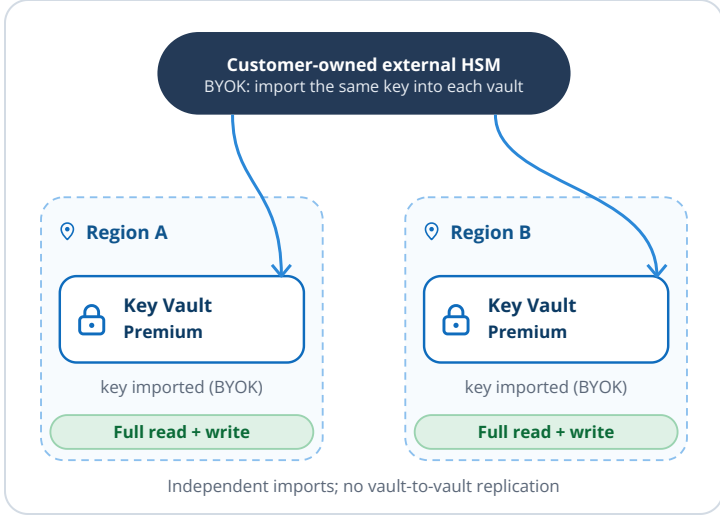
1 One Key Vault per region

CREATED IN	Key Vault Standard or Premium, one vault per workload per region.
OUTAGE	Each region works on its own vault, full read and write.
BEST FOR	Workloads whose services can use a separate key in each region.
WATCH FOR	Services that geo-replicate cannot use two vaults. Storage with GRS uses one key from one vault for both copies, so a vault outage affects both.



2 One Premium vault per region + external HSM

CREATED IN	Imported from your external HSM (BYOK) into a Premium vault in each region.
OUTAGE	Each region works on its own vault, full read and write.
BEST FOR	You must own and hold the master key, or meet BYOK rules, and want to control recovery time.
WATCH FOR	Same service limits as pattern 1, and extra cost.



CHECK BEFORE YOU PICK

Confirm that every service using the key supports a vault per region. Standard does not support BYOK; check supported HSMs for Premium. Restoring a vault backup works only within the same subscription and geography.

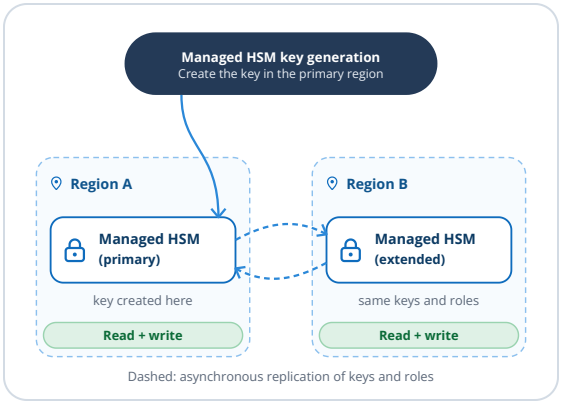
SERVICE GUIDES · KEY VAULT · KEYS (2 OF 2)

Keys: Managed HSM patterns

Patterns 3 to 5 use Managed HSM. Use two regions with replication, or keep the key in your own HSM.

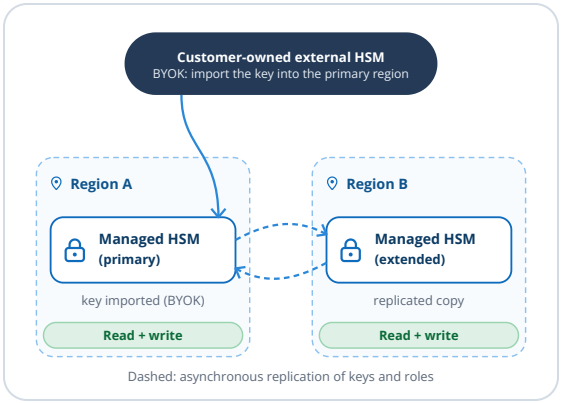
3 Managed HSM with geo-replication

CREATED IN	Managed HSM with multi-region replication.
OUTAGE	Reads and writes stay available in the healthy region.
BEST FOR	Own and hold the master key; mission-critical with customer-controlled recovery.
WATCH FOR	Cost. Changes replicate within 6 minutes; the last few may be lost.



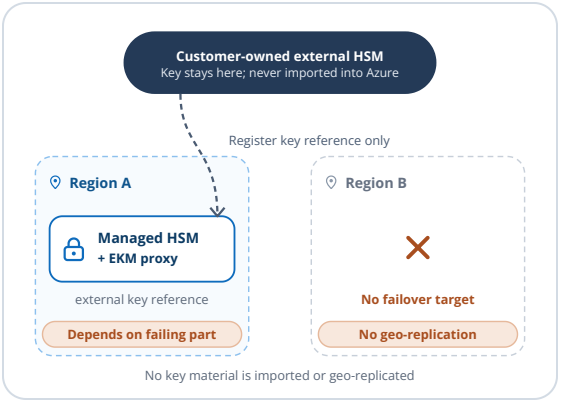
4 Managed HSM geo-replication + external HSM

CREATED IN	Imported from your external HSM, then replicated.
OUTAGE	Reads and writes stay available in the healthy region.
BEST FOR	Same as pattern 3, with your HSM as the key origin.
WATCH FOR	Cost. Same replication window as pattern 3.



5 Managed HSM without geo-replication + EKM

CREATED IN	Key never leaves your HSM.
OUTAGE	Depends on the failing part: Managed HSM, EKM proxy, your HSM, or the region. No failover target, so wait for restore.
BEST FOR	You must own and hold the key outside Microsoft.
WATCH FOR	No end-to-end SLA. Failures in your HSM can interrupt Azure workloads and do not count against their SLAs.

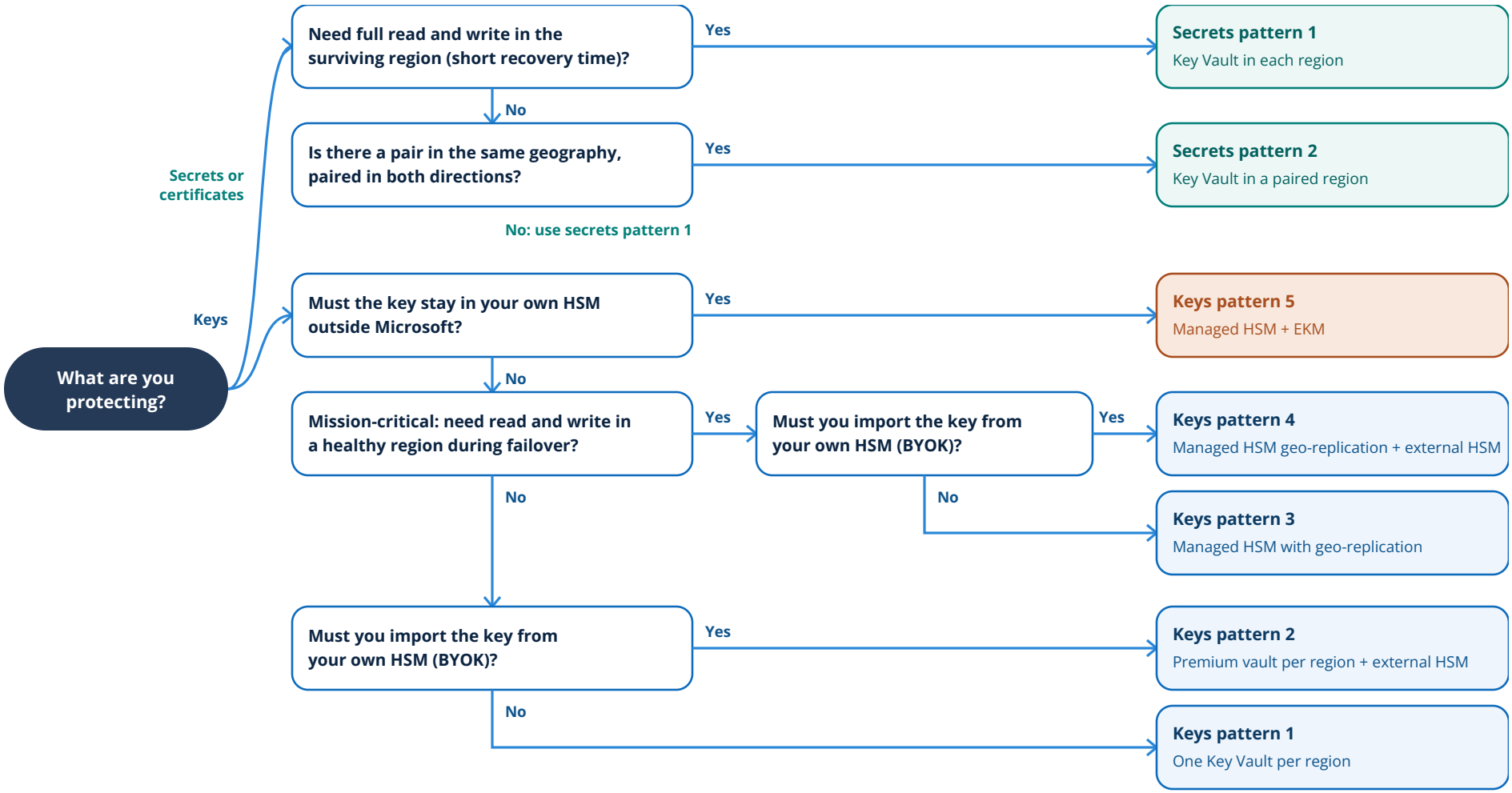


Managed HSM replicates to one extended region. Both regions are active and Azure Traffic Manager routes to the closest one; avoid caching DNS longer than the TTL. EKM adds network round trips to your proxy and HSM on every key operation. Recreate the external key reference instead of restoring a backup.

SERVICE GUIDES · KEY VAULT · DECISION TREE

Which Key Vault pattern fits?

Answer from the left. Each answer ends at one pattern from the previous pages. Validate current service behavior before you commit.



REFERENCE LINKS · MICROSOFT LEARN

- Reliability in Azure Key Vault
- Azure Key Vault availability and redundancy
- Azure paired regions
- Regions that support availability zones
- Managed HSM multi-region replication
- Managed HSM overview
- Supported HSMs for BYOK
- Services that support customer-managed keys
- How to choose the right key management solution
- Key Vault and Managed HSM SLAs

SERVICE GUIDES · PRIVATE DNS ZONES

Private DNS zones for multi-region private endpoints

Private DNS zones are global resources, so adding a region doesn't require new zones. Most workloads can continue using the existing landing zone's centrally managed `privatelink.*` zones. Resources with **private endpoints in more than one region** need additional DNS planning.

This page covers centralized DNS designs that share private DNS zones across regions. Decentralized designs, such as isolated spokes with their own zones, are outside its scope.

Why resources need private endpoints in multiple regions

A private endpoint is deployed in a virtual network and belongs to that network's region. Clients in other regions need routing to reach it. A resource may need private endpoints in multiple regions for:

- **Disaster recovery.** The service, the application, or both can fail over to another region. A private endpoint in the recovery region lets clients there reach the service without depending on the primary region.
- **Active/active.** The application runs in both regions simultaneously. Each region uses a local private endpoint to keep traffic local.

Why a single global zone can be a problem

When private endpoints for the same resource register the same fully qualified domain name (FQDN) in a shared private DNS zone:

- The zone holds a single A record for that FQDN, pointing to the private endpoint registered last.
- Clients in both regions receive the same IP address, so clients in the other region use cross-region connectivity.
- If the selected endpoint's region becomes unavailable, clients can't reach the resource until the record is updated to point to the surviving endpoint, even if the resource itself is healthy.

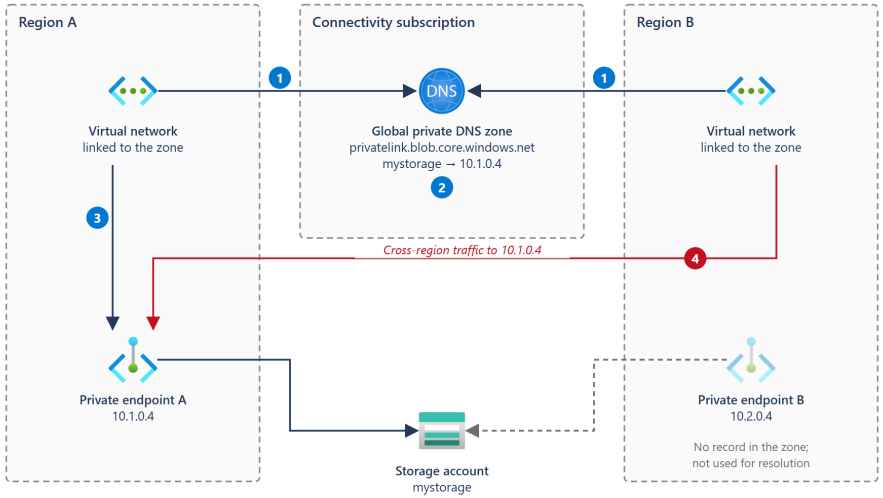


FIGURE 27 A shared global zone resolves the storage account to its private endpoint in Region A. Clients in Region B use cross-region connectivity.

1. Clients in both regions use the same global private DNS zone.
2. The record for `mystorage` points to the private endpoint in Region A.
3. Clients in Region A use their local private endpoint.
4. Clients in Region B receive the same IP address and use cross-region connectivity. Their local private endpoint isn't used. If Region A becomes unavailable, the record must be updated before clients can reach the resource.

   There is no conflict when the primary and secondary instances have different DNS names and each has a private endpoint only in its own region. If both instances also have private endpoints in the other region, each name has two endpoints competing for the same DNS record.

SERVICE GUIDES · PRIVATE DNS ZONES

Choose an approach

For a resource with private endpoints in multiple regions, choose one of two approaches:

- **Use a global zone and update the record during failover.** The A record normally points to the private endpoint in the primary region. During failover, update it to the secondary region's private endpoint, manually or through automation in the disaster recovery runbook.
- **Use regional zones.** For each required zone name, such as `privatelink.blob.core.windows.net`, create one private DNS zone per region. Link each zone only to that region's virtual networks. These zones are still global Azure resources; "regional" describes their use and virtual network links. Each zone resolves the FQDN to the local private endpoint, so no DNS record change is needed during failover.

1. Clients use the zone linked to their region's virtual networks.
2. Each zone resolves the resource name to the local private endpoint.
3. Clients access the resource through that endpoint without a DNS record change during failover.

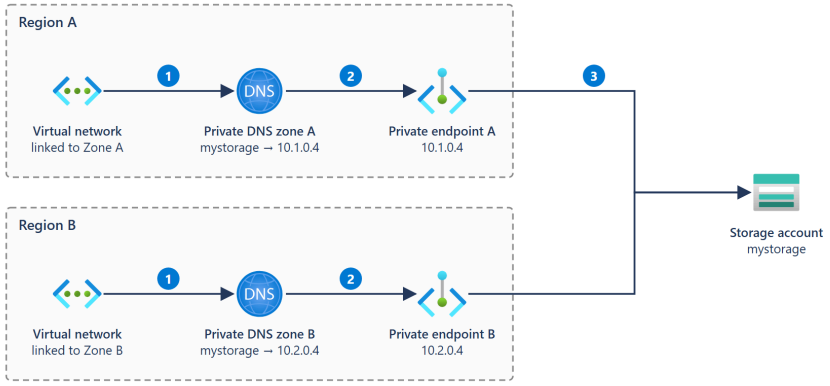


FIGURE 28 Separate zones with the same name provide a local private endpoint address in each region.

CONSIDERATION	GLOBAL ZONES	REGIONAL ZONES
Design	One zone for each required `privatelink.*` name, shared across regions	A separate zone with the same name for each region, linked only to that region
A records per FQDN	One	One per region
Resource with private endpoints in two regions	Clients use one endpoint. Failover requires a record update.	Each region uses its local endpoint. No DNS record update is needed during failover.
Operations	One set of zones. Azure landing zone `DeployIfNotExists` policies work unchanged.	Separate sets of zones, with records and policy assignments managed per region.
Cross-region access	All regions using the shared zone can resolve the resource.	Each region can resolve only the records in its zone.
On-premises	Resolvers using the shared zone return the same answer.	The answer depends on which regional resolver receives the query.

Use regional zones when a resource needs a private endpoint in each region for disaster recovery or active/active operation. A global zone is an alternative if cross-region connectivity is available, latency is acceptable, and DNS record updates during failover meet the recovery time objective (RTO).

SERVICE GUIDES · PRIVATE DNS ZONES

Updating the record during failover

With a global zone, DNS record updates become part of the failover process. This approach is suitable when:

- The RTO allows for a manual or automated DNS update.
- Cross-region routing lets clients reach the active private endpoint during normal operation, with acceptable latency.

Include the following in your recovery plan:

- **Update DNS during both failover and failback.** Document the update in each affected resource's runbook, automate it where possible, and test it.
- **Account for DNS caching.** Clients can continue using the previous IP address until the record's time to live (TTL) expires. Choose a TTL that supports the RTO.

Disaster recovery examples by service

SERVICE	DISASTER RECOVERY MODEL	PRIVATE ENDPOINTS FOR DISASTER RECOVERY	ZONE
Azure Storage (GRS/GZRS)	Account failover; the account name stays the same.	One per region, so clients use a local private endpoint after failover.	`privatelink.blob.core.windows.net` and zones for other subresources in use
Service Bus Premium and Event Hubs	**Geo-Replication:** a single namespace that can fail over between regions. **Geo-Disaster Recovery:** two namespaces, one in each region.	Create a private endpoint for each namespace in both regions: one endpoint per region for Geo-Replication, and two per region for Geo-Disaster Recovery.	`privatelink.servicebus.windows.net`
Azure SQL Database	A failover group has primary and secondary logical servers in different regions. The listener name follows the primary.	Create a private endpoint for each server in its own region. Both servers can share one private DNS zone.	`privatelink.database.windows.net`

The deciding factor is whether the same FQDN maps to private endpoints in multiple regions, not how many service instances exist. If it does, use regional zones or plan for DNS updates during failover. Different resource names can share a global zone without this conflict.

NOTE

Check each service's current documentation for its DNS behavior and private endpoint requirements.

SERVICE GUIDES · PRIVATE DNS ZONES

Considerations for regional zones

- **Choose the pattern per zone name, not per resource.** A virtual network can link to only one zone with a given name. If you use regional zones for `privatelink.blob.core.windows.net`, all storage account names under that zone are resolved through the zone linked to the client's network.
- **Include every record that clients need.** Region B's zone doesn't automatically contain records for private endpoints in Region A. Either add those records to Region B's zone and provide cross-region routing, or create local private endpoints in Region B and register them in its zone.
- **Target the correct zone in policy and automation.** The `DeployIfNotExists` policy takes a private DNS zone ID as a parameter. Assign it per region with the corresponding zone ID.
- **Use separate resource groups.** A resource group can't contain two zones with the same name.
- **Check on-premises DNS forwarding.** On-premises clients receive the answer from whichever regional resolver they query. Verify forwarding and resolution during both normal operation and failover.

Reference links

- Private Link and DNS integration at scale (Cloud Adoption Framework)
- Azure Service Bus Geo-Replication: private endpoints
- Azure Service Bus Geo-Disaster Recovery: private endpoints
- Azure Event Hubs geo-replication: private endpoints
- Azure Event Hubs geo-disaster recovery: private endpoints
- Failover considerations for storage accounts with private endpoints
- Configure a failover group for Azure SQL Database: use Private Link
- Multi-region use of Azure Private Link

RESOURCES · GETTING STARTED

Get started with a multi-region platform

Use this sequence to establish the first additional regional options. Start with one candidate region and the workloads that need it. The same process scales to a broader portfolio, and later regions reuse the decisions made for the first.

01 Name the drivers

Identify what is prompting the discussion — business growth, regulation, service or SKU availability, resiliency, a datacenter or connectivity event, cost, or another measurable requirement.

Drivers

02 Understand the concepts

Learn how a multi-region platform differs from a multi-region workload, the four connectivity profiles a region can provide, and the four Workload Landing Zone archetypes that describe what applications need. You make no decisions here. You learn the terms the next steps use.

Platform and workload · Connectivity profiles · Archetypes

03 Shortlist and qualify candidate regions

Work through the four qualification checks. Start with the two that need no technical assessment, business and geography and data and compliance, and remove regions that cannot satisfy mandatory conditions. For the regions that remain, identify workload requirements and dependencies, then validate regional capability and availability. Record each option as excluded, candidate, conditional, or qualified for the defined requirements, with evidence and revalidation triggers.

Checks 1–4 · Qualification tree · Assessment outcome

04 Select the regional design

For each qualified region, select the most efficient connectivity profile that meets the requirements, decide where each shared service is provided, and confirm whether the hybrid connectivity you already have is enough. Write the three choices into the regional design record, and agree who owns each decision.

Three choices · Profile tree · Design record

05 Prepare the region

Build what the record says, through the automation you already run. Read only the page for the profile you selected: Disconnected Spokes, Remote Hub Connected, Minimal Regional Hub, or Full Regional Hub. Use the platform readiness checklist to confirm the region is ready before the first workload arrives.

Profile pages · Readiness checklist

06 Place the first workload

Apply the workload-placement process using current evidence. Revalidate regional capabilities at decision time, govern the placement and resiliency decisions, and feed newly discovered requirements or platform gaps back into regional qualification.

Placement flow

WHAT SUCCESS LOOKS LIKE

Being able to qualify a regional option, enable only the platform capabilities it requires, and onboard workloads through a repeatable governed process.

THE FRAMEWORK IN USE

Customer scenarios

How the framework moves from a business or technical driver to a governed regional and workload decision. Illustrative, not customer-specific recommendations.

1 Regulated bank adopts a new in-country region

Residency · nonpaired region

A financial-services organization operates workloads outside its home country. A new Azure region opens in-country while regulatory or business requirements create a need for certain data or services to remain there.

DRIVER	DECIDING CHECKS	QUALIFICATION	ARCHETYPE	CONNECTIVITY PROFILE	PLACEMENT
Data residency and regulatory requirements; new in-country Azure region.	Check 2 confirms mandatory location requirements. Check 3 identifies affected workloads and dependencies. Check 4 validates required services, zones, access, and constraints.	Qualified for defined Hybrid-Connected requirements; other portfolio workloads remain candidates until dependencies are understood.	Hybrid-Connected Applications.	Minimal Regional Hub where selected local capabilities are required; existing hybrid connectivity reused where reachability and resiliency permit.	In-scope workloads relocate selectively; others remain in place. Recovery architecture stays workload-specific.

Takeaway. A new in-country region can be qualified for defined workload requirements without forcing portfolio-wide relocation. If the new region is nonpaired, design the recovery path explicitly (Resources).

2 AI team requires a specific model or GPU SKU

Specific capacity · net-new

A product team requires an AI model, GPU SKU, or related Azure service that is unavailable in the organization's existing regions or constrained by quota.

DRIVER	DECIDING CHECKS	QUALIFICATION	ARCHETYPE	CONNECTIVITY PROFILE	PLACEMENT
AI model, service, SKU, or quota availability.	Check 1 confirms the business requirement. Check 2 validates data and compliance constraints. Check 4 validates current model, service, SKU, zone, access, and quota support.	Qualified for the defined AI workload requirements; conditional for requirements that depend on private connectivity until that design is known and validated.	Isolated Cloud-Native or AI where self-contained; Connected Cloud-Native or AI where private enterprise dependencies are required.	Disconnected Spokes where the workload is self-contained; a connected profile only where actual dependencies justify it.	Net-new deployment. Revalidate model, SKU, service, regional access, and quota at decision time.

Takeaway. Qualify against the requirements that exist today; do not introduce enterprise connectivity or regional platform capabilities before a workload requirement justifies them.

CUSTOMER SCENARIOS, CONTINUED

3 Acquisition brings an estate in another geography

Portfolio · integration

An acquisition introduces a portfolio of interdependent applications, shared data, and local users in a geography where the organization has little or no existing Azure platform presence.

DRIVER	DECIDING CHECKS	QUALIFICATION	ARCHETYPE	CONNECTIVITY PROFILE	PLACEMENT
Business expansion through acquisition.	Check 1 confirms the business need. Check 2 evaluates local data obligations. Check 3 maps portfolio-level dependencies, shared services, and connectivity requirements.	Candidate while material dependencies remain unknown; qualified against defined portfolio requirements once discovery is complete.	Interconnected Application Portfolios.	Remote Hub Connected where cross-region dependencies are acceptable; Minimal Regional Hub where east-west traffic, shared services, latency, or independence justify it.	Workloads remain in place initially and relocate selectively as business and integration requirements become known.

Takeaway. Aggregate portfolio dependencies determine the appropriate regional connectivity profile; application count or acquisition status does not.

4 Datacenter exit accelerates migration

Datacenter event · relocation

A colocation contract is ending. Hybrid-connected applications must leave the datacenter, and the organization asks whether the event also requires adoption of another Azure region.

DRIVER	DECIDING CHECKS	QUALIFICATION	ARCHETYPE	CONNECTIVITY PROFILE	PLACEMENT
Datacenter and connectivity event.	Check 3 identifies what must move together. Part 3 validates hybrid connectivity, datacenter and user geography, reachability, resiliency, and whether existing connectivity suffices.	The existing target region remains qualified for the identified requirements; an additional region remains a candidate for future use.	Hybrid-Connected Applications.	No new profile. Continue to use the existing strategic geographic hub where it satisfies requirements; no new regional hub solely because another region is available.	Most workloads land in the existing qualified region. Revisit the additional option when requirements change.

Takeaway. Remaining in an existing qualified region is a valid outcome. An additional region can remain an option rather than automatically becoming a deployment project.

RESOURCES · COMMON MISCONCEPTIONS

Common misconceptions

Multi-region programs often become more complex than necessary when teams start from assumptions instead of workload and platform requirements. The corrections cover qualification, platform capabilities and connectivity, workload placement, service guides, and Resources.

✕ "Adding a region means another landing-zone project, with its own governance model."

✓ The landing zone you already run is reused: governance, identity, policy, automation, and the operating model stay the same. A new region adds regional configuration and only the connectivity profile its workloads need.

✕ "We must choose the perfect regions up front."

✓ Regional requirements and Azure capabilities change. Design the platform so additional qualified options can be introduced without redesigning the broader operating model.

✕ "Qualified means approved for any workload."

✓ Qualification is scoped to defined requirements and evidence at a point in time. Each workload revalidates regional suitability during onboarding.

✕ "Every new region needs a Full Regional Hub."

✓ Select the **most efficient profile** that satisfies the identified requirements. Profiles are not maturity stages; any profile can remain the long-term target while it continues to meet those requirements.

✕ "Another Azure region requires another ExpressRoute circuit."

✓ An additional region does not by itself require another circuit. Additional hybrid connectivity should be introduced where reachability, diversity, failure-domain, latency, routing, or bandwidth requirements justify it.

✕ "Enabling a region means migrating workloads."

✓ Regional qualification and platform readiness are not workload placement. Remaining in the existing region can be the appropriate outcome.

✕ "A paired region is our disaster recovery."

✓ Region pairing supports specific Azure platform and service behaviors; it does not automatically provide workload recovery. Recovery must still be designed and validated per workload.

✕ "A multi-region platform means active-active."

✓ The platform creates regional options. Whether a workload is single-region, primary/recovery, or active across regions is a separate workload-specific resiliency decision.

Each correction maps to the framework: profiles (Part 1), qualification (Part 2), zones and pairing (check 4 and Resources), the regional design, including hybrid connectivity (Part 3), preparing the region (Part 4), placement (Part 5), and the operating model (Framework overview).

RESOURCES · REGION PAIRING

Paired and nonpaired regions

Azure supports resilient architectures using paired regions, nonpaired regions, or combinations of both; the appropriate approach depends on workload requirements and the capabilities of the Azure services involved.

Azure paired regions

PLATFORM-ASSISTED REGIONAL RELATIONSHIP

- Microsoft defines region pairs, usually within the same Azure geography, although asymmetric and cross-geography exceptions exist.
- Some Azure services use the paired region for built-in geo-replication or geo-redundancy — for example, geo-redundant storage, Key Vault, and Azure Backup Cross Region Restore.
- Azure uses region pairs for sequential platform updates and recovery prioritization.
- Deploying into both members of a pair does **not** automatically provide workload high availability, disaster recovery, or failover.

Consider: built-in service behavior can constrain the secondary region to the Azure-defined pair. Some paired regions have restricted access, and individual services can have different regional capabilities. Validate current service behavior before selecting a recovery architecture.

Nonpaired or region-of-choice designs

ARCHITECTURE-LED RECOVERY

- Many newer Azure regions are nonpaired and use availability zones as an important part of their intra-region resiliency model.
- A multi-region workload can use a nonpaired or customer-selected secondary region where the required Azure services support that regional combination — through Azure service-based or ISV-based disaster recovery.
- Provides flexibility for residency, sovereignty, latency, service availability, or business-continuity requirements.
- The recovery architecture, replication method, dependencies, and failover process must be explicitly designed and validated per workload.

Pairing can provide useful platform and service capabilities; it does not replace a tested workload recovery design.

> "Deploying resources to a region in a pair doesn't automatically make them more resilient, nor does it provide automatic high availability, disaster recovery capabilities, or failover."
>
> MICROSOFT LEARN · AZURE REGION PAIRS AND NONPAIRED REGIONS

WHAT TO RECORD DURING QUALIFICATION

Assess availability zones, region pairing, and service-specific cross-region behavior separately (Figure 16). For each regional combination that workloads are expected to use, record:

- Whether each region is paired, which region it is paired with, and whether access to the paired region is restricted.
- For each service the workloads depend on, whether its cross-region capability is tied to the paired region or supports a region of choice.
- Availability-zone support in each region.
- The intended recovery region for the defined workload requirements, and who validates the recovery path.

RESOURCES · RESILIENCY SCENARIOS

Multi-region resiliency scenarios

These examples show how service-specific regional behavior changes workload recovery design. They complement the broader customer scenarios earlier in this paper.

A workload in a nonpaired region requires DR

A workload operates in a zone-enabled region without a pair and must recover in another Azure region.

STORAGE	A supported region-of-choice pattern such as object replication where suitable, or application-level replication where required (for example, Data Lake Storage).
COMPUTE	Azure Site Recovery for applicable VMs, or redeploy stateless compute from infrastructure as code.
KEY VAULT	Pre-provision the regional vault; runtime secrets, certificates, and any key strategy available before recovery.
BACKUP	Place backup and restore capabilities deliberately for the intended recovery region.

Takeaway. A nonpaired region can participate in a resilient multi-region architecture, but each workload layer needs an explicit recovery design.

"We're paired, so we have DR"

Data is geo-replicated to the paired region, but the complete application has never been failed over and tested.

SYMPTOM	Data is recoverable, but the application cannot operate in the secondary region.
POSSIBLE CAUSES	Key Vault or another runtime dependency unavailable (failover not yet occurred, or read-only); DNS or private-endpoint configuration incomplete; compute or shared services not recoverable; dependencies missing from the recovery design.
CORRECTION	Treat data, identity, secrets and keys, DNS, networking, shared services, and compute as workload dependencies. Validate end-to-end recovery against RTO and RPO.

Takeaway. Data recovery is not application recovery, and region pairing does not replace a tested workload recovery plan.

Customer-managed keys across two regions

A workload uses customer-managed encryption keys (for example, database TDE) and must operate or recover across two regions.

CONSTRAINT	Key material has different recovery and replication characteristics from secrets and certificates; it cannot be assumed to be recreated by the pipeline.
OPTIONS	Key Vault behavior for the selected regions (paired-region replication where available); backup and restore within the same subscription and geography; Azure Managed HSM multi-region replication; or another workload-specific design.
VALIDATE	Key availability, access, private connectivity, permissions, and application behavior in the secondary region before the recovery path is considered complete.

Takeaway. Design key availability explicitly. For some workloads, key management determines which recovery regions are viable.

FIGURE 29 **Resiliency scenarios.** Representative situations, not drawn from a specific customer. Confirm current service capabilities on Microsoft Learn at decision time.

CONCLUSION

Design for continuing regional choice

No organization can predict which Azure regions will offer the best mix of services, regulatory fit, business proximity, and workload suitability several years from now. **Design the platform so that perfect prediction is unnecessary.**

This framework extends Azure landing-zone principles rather than replacing them. Regional adoption, workload placement, platform expansion, and resiliency can evolve without redesigning the broader platform for every new region.

Workload-led placement

Qualified regional options, Workload Landing Zone archetypes, and current workload requirements allow applications to use the regional and platform capabilities appropriate to their needs.

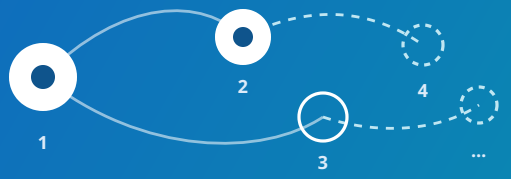
Adaptable regional platform

Strategic geographic hubs, resilient hybrid connectivity, and regional connectivity profiles distribute capabilities according to workload requirements — without an identical footprint in every region.

Consistent operating model

Keep the landing-zone guardrails and operating standards stable across the Azure estate.

The goal is not to select the perfect two regions. **It is to make additional regions significantly easier to adopt when business and workload requirements justify them.**



REFERENCES AND FURTHER READING

Microsoft Learn and Azure resources cited in this paper

MULTI-REGION WORKLOADS

Using availability zones and regions — Azure Well-Architected Framework
Deployment approaches, including multi-region deployments.
learn.microsoft.com/azure/well-architected/design-guides/regions-availability-zones

CHECK 1 · BUSINESS AND GEOGRAPHY

Select Azure regions
Aligning regional choices with user locations, business requirements, and deployment considerations.
learn.microsoft.com/azure/cloud-adoption-framework/ready/azure-setup-guide/regions

Azure geographies
Azure's geographic footprint and locations relevant to markets and operations.
azure.microsoft.com/explore/global-infrastructure/geographies

CHECK 2 · DATA AND COMPLIANCE

Data residency in Azure
Customer-data location, geographic boundaries, and service-specific exceptions.
azure.microsoft.com/explore/global-infrastructure/data-residency

Azure compliance documentation
Regional, industry-specific, and global compliance references.
learn.microsoft.com/azure/compliance

Reliability and sovereignty in Azure
Sovereignty and regional selection, replication destinations, encryption, and operational control.
learn.microsoft.com/azure/reliability/concept-reliability-sovereignty

ARCHETYPES AND CHECK 3 · WORKLOAD REQUIREMENTS

Architecture styles
N-tier, microservices, event-driven, and other styles.
learn.microsoft.com/azure/architecture/guide/architecture-styles

Evaluate a cloud workload for relocation
Dependency discovery, workload ownership, acceptable disruption, and target-region supportability.
learn.microsoft.com/azure/azure-resource-manager/management/relocate-evaluate

Relocate cloud workloads
The Azure-to-Azure relocation process; assessment distinct from migration and cutover.
learn.microsoft.com/azure/azure-resource-manager/management/relocate-index

CHECK 4 · REGIONAL CAPABILITY AND AVAILABILITY

Azure region pairs and nonpaired regions
learn.microsoft.com/azure/reliability/regions-paired

Azure services that support multiple regions
learn.microsoft.com/azure/reliability/regions-multiregion-support

Azure services that support availability zones
learn.microsoft.com/azure/reliability/availability-zones-service-support

Zonal resources and zone resiliency
learn.microsoft.com/azure/reliability/availability-zones-zonal-resource-resiliency

Enable zone resiliency for Azure workloads
learn.microsoft.com/azure/reliability/availability-zones-enable-zone-resiliency

PART 3 · SELECT THE REGIONAL DESIGN

Azure landing zone design areas and conceptual architecture
Identity, security, governance, management, connectivity, and deployment automation.
learn.microsoft.com/azure/cloud-adoption-framework/ready/landing-zone/design-areas

Designing for disaster recovery with ExpressRoute private peering
Peering-location diversity, redundant paths, and failover.
learn.microsoft.com/azure/expressroute/designing-for-disaster-recovery-with-expressroute-privatepeering

Network topology and connectivity
Connectivity between workloads, shared services, and the broader platform.
learn.microsoft.com/azure/cloud-adoption-framework/ready/landing-zone/design-area/network-topology-and-connectivity

Define an Azure network topology
Topology approaches based on cross-region connectivity and routing requirements.
learn.microsoft.com/azure/cloud-adoption-framework/ready/azure-best-practices/define-an-azure-network-topology

WORKLOAD DESIGN AND RESILIENCE

Architecture strategies for designing for redundancy
Multi-region deployment models, including active-active and active-passive.
learn.microsoft.com/azure/well-architected/reliability/highly-available-multi-region-design

Design methodology for mission-critical workloads
The full multi-region active design for the most demanding workloads.
learn.microsoft.com/azure/well-architected/mission-critical/mission-critical-design-methodology

Multi-region solutions in nonpaired regions
Service-by-service guidance for regions without a pair.
learn.microsoft.com/azure/reliability/cross-region-replication-azure-no-pair

Azure reliability service guides
Regional behavior for each Azure service.
learn.microsoft.com/azure/reliability/overview-reliability-guidance

REFERENCES AND FURTHER READING · CONTINUED

PART 3 · PLACE SHARED SERVICES

Design a Log Analytics workspace architecture
When a single workspace is enough and when to add one per region.
learn.microsoft.com/azure/azure-monitor/logs/workspace-design

Private Link and DNS integration at scale
Central private DNS zones and the policies that keep their records current.
learn.microsoft.com/azure/cloud-adoption-framework/ready/azure-best-practices/private-link-and-dns-integration-at-scale

Hybrid identity in Azure landing zones
Where to place domain controllers, including across regions.
learn.microsoft.com/azure/cloud-adoption-framework/ready/landing-zone/design-area/identity-access-active-directory-hybrid-identity

Azure Backup support matrix
Vault and region requirements.
learn.microsoft.com/azure/backup/backup-support-matrix

PART 3 · CONFIRM HYBRID CONNECTIVITY

Connectivity to Azure
ExpressRoute and VPN recommendations for landing zones, including circuits from different peering locations.
learn.microsoft.com/azure/cloud-adoption-framework/ready/azure-best-practices/connectivity-to-azure

Use a VPN or ExpressRoute gateway in a different region
Gateway transit over global virtual network peering, with its requirements and limitations.
learn.microsoft.com/azure/vpn-gateway/vpn-gateway-different-region

Traditional Azure networking topology
A hub per region, how hubs share ExpressRoute circuits, and cross-region inspection.
learn.microsoft.com/azure/cloud-adoption-framework/ready/azure-best-practices/traditional-azure-networking-topology

Global transit network architecture and Virtual WAN
Cross-region virtual network connections and hub-to-hub transit.
learn.microsoft.com/azure/virtual-wan/virtual-wan-global-transit-network-architecture

PART 4 · PREPARE THE REGION

Landing zone regions: Add a new region to an existing landing zone
Regional considerations and steps to add a region, area by area, including a new hub.
learn.microsoft.com/azure/cloud-adoption-framework/ready/considerations/regions#add-a-new-region-to-an-existing-landing-zone

Management groups
Why the structure doesn't change for more regions.
learn.microsoft.com/azure/cloud-adoption-framework/ready/landing-zone/design-area/resource-org-management-groups

Subscription vending
Automated provisioning of landing zones for workloads.
learn.microsoft.com/azure/cloud-adoption-framework/ready/landing-zone/design-area/subscription-vending

Plan for inbound and outbound internet connectivity
learn.microsoft.com/azure/cloud-adoption-framework/ready/azure-best-practices/plan-for-inbound-and-outbound-internet-connectivity

Default outbound access in Azure
Private subnets and the explicit outbound methods.
learn.microsoft.com/azure/virtual-network/ip-services/default-outbound-access

DNS for on-premises and Azure resources
learn.microsoft.com/azure/cloud-adoption-framework/ready/azure-best-practices/dns-for-on-premises-and-azure-resources

Virtual network peering
learn.microsoft.com/azure/virtual-network/virtual-network-peering-overview

Virtual WAN network topology
A virtual hub per region, secured hubs, and routing intent.
learn.microsoft.com/azure/cloud-adoption-framework/ready/azure-best-practices/virtual-wan-network-topology

Hub-spoke network topology in Azure
A hub per region, what a hub hosts, and connectivity between spokes.
learn.microsoft.com/azure/architecture/networking/architecture/hub-spoke

SERVICE GUIDES FOR MULTI REGION

Object replication for block blobs
Supported blob types, prerequisites, and limits for region-of-choice replication.
learn.microsoft.com/azure/storage/blobs/object-replication-overview

Azure Storage redundancy
LRS, ZRS, GRS, and GZRS options and failover.
learn.microsoft.com/azure/storage/common/storage-redundancy

Restore Azure VMs — Cross Region Restore
Restoring VMs to the paired secondary region.
learn.microsoft.com/azure/backup/backup-azure-arm-restore-vms

Azure Site Recovery support matrix
Region support for Azure-to-Azure disaster recovery, including restricted regions.
learn.microsoft.com/azure/site-recovery/azure-to-azure-support-matrix

Azure Key Vault availability and redundancy
Replication, Microsoft-managed failover, and backup and restore limits.
learn.microsoft.com/azure/key-vault/general/disaster-recovery-guidance

Reliability in Azure Key Vault
Custom multi-region solutions for resiliency.
learn.microsoft.com/azure/reliability/reliability-key-vault

Managed HSM multi-region replication
Extending a Managed HSM pool to an additional region.
learn.microsoft.com/azure/key-vault/managed-hsm/multi-region-replication

TOOLS

Multi-Region Platform website
Read the guidance and use the region planning tool.
azure.github.io/multi-region-platform

Multi-region planner
Compare regions, record evidence and designs, export Excel, or save a JSON plan.
azure.github.io/multi-region-platform/region-planner.html

A NOTE ON THE FIGURES

Region names, service placements, and estates shown in the figures are illustrative. They explain the framework; they do not recommend specific Azure regions or topologies.

SERVICE CAPABILITIES CHANGE

Regional service behavior in the service guides reflects Microsoft Learn at the time of writing. Confirm current capabilities at decision time.

Microsoft

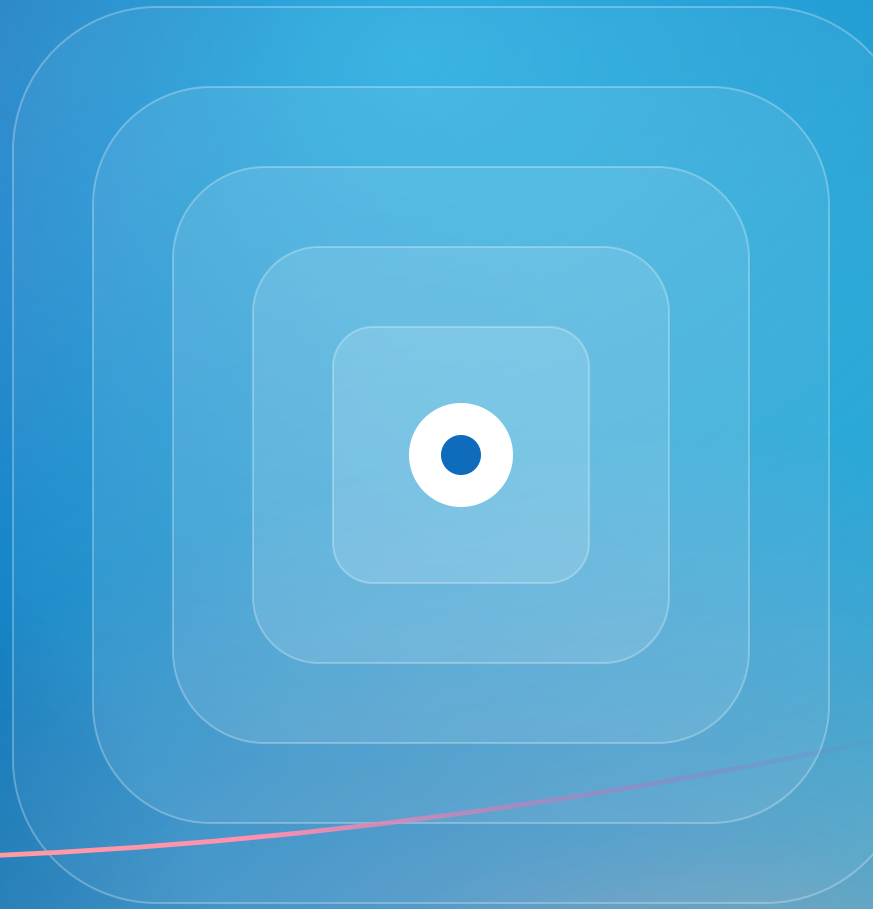
Whitepaper

The adaptable **multi-region** Azure platform

Extending Azure landing zones for adaptable regional placement through Regional Platform Connectivity Profiles, Workload Landing Zone Archetypes, and Regional Qualification.

Multi-Region Platform Whitepaper · October 2026

Read online	azure.github.io/multi-region-platform
Multi-region planner	azure.github.io/multi-region-platform/region-planner.html



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